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Overcoming Breakdowns in Customer-Chatbot Interaction: Design and Impact of Collaborative Repair Strategies[1]

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When chatbots are deployed to automate customer service, it is nearly inevitable that situations will arise in which they struggle to understand customer requests. Unfortunately, the onus of resolving such conversational breakdowns tends to fall on either the customer or the chatbot alone, turning customer-chatbot interaction into a frustrating and often unsuccessful guessing game. Despite indications that customers would be open to collaboration, we know little about repair strategies that involve the customer and chatbot working together to resolve breakdowns. Our research addresses this gap by investigating the design and impact of collaborative repair strategies in customer-chatbot interaction. Drawing upon an integration of the theory of least collaborative effort with research on human-machine communication and customer service chatbots, we propose a novel repair strategy design; we instantiated it in the chatbot of a large insurance company and conducted a naturalistic summative evaluation through a randomized field experiment. Overall, our results suggest that a collaborative repair strategy can lead to more breakdowns being resolved and mitigate the negative impacts of breakdowns on key customer outcomes. Our research offers a new way of thinking about customer-AI service interactions by shifting the narrative from confrontation to collaboration, extends the theory of least collaborative effort by integrating the perspective of customer-chatbot interaction, and provides in-depth insights into breakdown and repair in real-world conversations between customers and chatbots.

Keywords: Customer service chatbot, conversational breakdown, repair strategy, service failure, human-AI interaction, theory of least collaborative effort, design science research, field experiment

Introduction

More and more organizations are deploying artificial intelligence (AI)-based chatbots to provide timely, efficient, and cost-effective customer service. Accordingly, the global chatbot market has been growing rapidly, with a projected value of $22.9 billion by 2030 (Reports and Data, 2022). However, despite advances in AI, current chatbots often manage to irritate customers (Brendel et al., 2023). According to market research, 80% of U.S. customers find using chatbots frustrating, and 72% even consider them a waste of time (UJET, 2022).

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One of the most frustrating situations for customers is when a chatbot fails to understand their requests. These *conversational breakdowns* disrupt the interaction flow, cause customers to abandon the chatbot, and lead to dissatisfaction and negative word of mouth (Diederich et al., 2021; Han et al., 2021; Seeger & Heinzl, 2021). Unfortunately, the complexity and ambiguity of natural human language make it difficult to prevent breakdowns in customer-chatbot interaction from occurring altogether (Brendel et al., 2023; Han et al., 2021). Therefore, researchers and practitioners have begun to realize the importance of *repair strategies* that can resolve breakdowns and mitigate their negative impacts (Ashktorab et al., 2019; Benner et al., 2021).

A growing body of research at the intersection of the literature on service failure and chatbot design seeks to understand how breakdowns occur in customer-chatbot interaction and to evaluate repair strategies for resolving them. However, our review of this research suggests that prior studies have focused on repair strategies that tend to put the onus of resolving breakdowns on either the customer or the chatbot alone (see Appendix A). In the former case, chatbots typically ask customers to repeat or reformulate their request (e.g., “Sorry, I didn’t understand that. Please try again”). As a result, customers can easily find themselves in a frustrating process of trial and error (Li et al., 2020a; Rapp et al., 2023). In the latter case, chatbots try to provide the response that is most likely to address the customer’s request, usually without acknowledging that a breakdown has occurred. However, this can easily backfire if the chatbot is unable to accurately anticipate the customer’s information need (Han et al., 2022; Verne et al., 2022). Therefore, both strategies often turn the interaction into a guessing game, with the customer trying to guess what the chatbot can understand or the chatbot trying to guess what the customer meant. Although research has embraced the idea of human-AI collaboration in various domains (Anthony et al., 2023; Seeber et al., 2020), and there is some indication that customers would be willing to resolve a breakdown together with a chatbot (Rapp et al., 2023), we know little about designing *collaborative* repair strategies. Moreover, it is unclear whether such strategies would be more effective than existing repair strategies in resolving breakdowns, particularly in real-world settings where breakdowns occur naturally.

Our research addresses these knowledge gaps by investigating the design and impact of collaborative repair strategies in real-world customer-chatbot interaction. Following the design science research (DSR) paradigm (Hevner et al., 2004; Walls et al., 1992), we first proposed a novel design for collaborative repair strategies that involve both the customer and the chatbot in the process of resolving a breakdown. Our proposed design is grounded in the theory of least collaborative effort (TLCE; Clark & Wilkes-Gibbs, 1986), which we integrated with research on human-machine communication and customer service chatbots before using it as our kernel theory. We then instantiated our design in collaboration with InsurCo, a large international insurance company that offers a customer service chatbot for a number of its products. In a three-phase design process, we implemented our repair strategy using a combination of real-world interaction data (N = 5,668), machine learning approaches, expert feedback, and controlled online experiments (N = 579). Finally, we conducted a naturalistic summative evaluation through a randomized field experiment with InsurCo’s customers (N = 1,352), which suggests that a collaborative repair strategy can increase customers’ willingness to engage in repair, lead to more breakdowns being resolved, and mitigate the negative impacts of breakdowns on key customer outcomes.

This research makes the following contributions. First, it adds a new perspective to the literature on customer-AI service interactions by shifting the narrative from confrontation to collaboration. Whereas existing literature tends to implicitly portray customers and chatbots as adversaries, our research frames them as communication partners working together to resolve problems. This stimulates new ways of thinking, not only about the relationship between customers and chatbots in service settings but also about a potential shift of the power dynamics in human-AI collaboration as AI begins to take on more powerful roles (e.g., as leaders instead of assistants). Second, our research extends TLCE by integrating the perspective of customer-chatbot interaction. The resulting framework, which we call TLCE-bot, introduces three core principles that explicate the dynamics of breakdown and repair in this domain and expand the scope of the original theory. Third, our research adds to the growing literature on customer-chatbot interaction by offering insights into the nature of breakdowns and the impact of collaborative repair strategies in real-world settings. Finally, our findings provide actionable design prescriptions for practitioners developing or managing customer service chatbots and practical guidance for organizations seeking to improve human-AI collaboration in other domains.

This paper is structured as follows. First, we review existing research at the intersection of the literature on service failure and chatbot design to pinpoint the knowledge gaps motivating our work. Next, we develop a theory-driven design for collaborative repair strategies following the DSR paradigm. We then describe the instantiation of our proposed design at InsurCo and its evaluation through a field experiment. We conclude by discussing implications for research and practice and avenues for future research.

Related Literature

Our work is positioned at the intersection of two streams of literature: service failure and chatbot design. In the following, we briefly summarize the research in each of these streams before reviewing studies at their intersection, which are those most closely related to our work. We then highlight the knowledge gaps that we aim to address.

Service Failure

Over the past decades, researchers in information systems (IS) and marketing have studied the topic of failure in service interactions extensively. While they traditionally focused on failures in face-to-face interactions, the increasing prevalence of self-service applications has redirected their attention toward information technology (IT)-based failures (Scherer et al., 2015; Tsohou et al., 2020). In this context, service failures are understood as negative events that occur whenever a self-service application lacks the technological capabilities that customers require for accomplishing their transactional activities or objectives (Tan et al., 2016). As service failures are inevitable and undermine customer satisfaction, research has highlighted the importance of service recovery strategies (e.g., offering apologies or explanations) in addressing these failures and mitigating their negative impact (Smith et al., 1999; Tsohou et al., 2020). Thus far, most studies on IT-based service failure and recovery have focused on established self-service applications such as web portals and e-commerce websites (e.g., Tan et al., 2016; Tsohou et al., 2020). However, researchers have recently begun to investigate how AI-based self-service applications, such as service robots and chatbots, can not only cause service failures but also help resolve them (e.g., Cheng et al., 2022; Choi et al., 2021).

Chatbot Design

Owing to major advances in AI, more and more organizations are launching chatbots as online self-service applications designed to communicate with customers using natural language (Han et al., 2023). These chatbots—also called text-based conversational agents or conversational AI agents—use AI techniques, such as machine learning and natural language processing, to understand customer requests and generate responses (Schanke et al., 2021). Their widespread adoption has sparked substantial interest in chatbots among IS researchers (Diederich et al., 2022). A major focus of this growing research stream is the empirical investigation of how chatbot design and use influence customer perceptions and behaviors (e.g., Han et al., 2023; Schuetzler et al., 2020). One central point of debate is whether organizations should "humanize" their customer service chatbots and how this decision might impact customer experiences (Chandra et al., 2022; Crolic et al., 2022; Schanke et al., 2021). Studies in this stream also offer guidance for developing chatbots (e.g., Elshan et al., 2023) and innovative chatbot artifacts for specific contexts (e.g., border screening; Nunamaker et al., 2011). In general, this research highlights that despite technological advances, designing a chatbot remains challenging, and effective repair strategies are needed to address the inevitable failures occurring in chatbot interactions (Elshan et al., 2023; Schuetzler et al., 2021).

Chatbot Breakdown and Repair Strategies

Research on chatbot breakdown and repair strategies lies at the intersection of the two literature streams introduced above. The impetus for this research comes from the recognition that service failures also occur in the chatbot context and must be addressed to mitigate their negative consequences (de Sá Siqueira et al., 2024; Elshan et al., 2023; Haupt et al., 2023). The most common failure is when a chatbot is unable to understand a customer's request (Diederich et al., 2021; Han et al., 2021; Weiler et al., 2022); this situation is usually referred to as a conversational breakdown (Ashktorab et al., 2019; Benner et al., 2021). As breakdowns can disrupt the flow of interaction, cause frustration, and undermine customer satisfaction (Brendel et al., 2023; de Sá Siqueira et al., 2024; Seeger & Heinzl, 2021), repair strategies are needed to resolve breakdowns and mitigate their negative impacts during ongoing chatbot interactions (Benner et al., 2021; Braggaar et al., 2023; Haupt et al., 2023; Law et al., 2022).

Existing research on chatbot breakdown and repair strategies can be broadly classified into four main categories based on the primary focus: (1) breakdown exploration, (2) breakdown impact, (3) repair strategy exploration, and (4) repair strategy impact (see Appendix A for details). First, breakdown exploration studies employ inductive, data-driven approaches to understand the situations in which breakdowns occur. They typically explore datasets of real-world interactions between customers and a specific chatbot (Li et al., 2020a; Nguyen et al., 2021; Rapp et al., 2023). Studies in this category have observed that breakdowns in such interactions are common (Li et al., 2020a; Verne et al., 2022) and can have a range of underlying causes, including ambiguous, complex, or underspecified questions and out-of-scope requests (Rapp et al., 2023). When breakdowns occur, some customers abandon the chatbot immediately (Li et al., 2020a), while others try to resolve the breakdown on their own, often without success (Dippold, 2023; Nguyen et al., 2021). Taken together, these studies reveal the prevalence of breakdowns in real-world customer-chatbot interaction and the struggles customers face in attempting to resolve them, highlighting the need for effective repair strategies.

Second, breakdown impact studies empirically investigate how breakdowns affect customer perceptions and attitudes toward chatbots and the organizations employing them. Primarily using surveys and experiments, these studies reveal negative effects on a wide range of perceptions (e.g., usefulness, trust), emotions (e.g., frustration, anger), and downstream outcomes (e.g., satisfaction, word of mouth, chatbot adoption) (e.g., Brendel et al., 2023; Diederich et al., 2021; Følstad et al., 2024; Law et al., 2022; Seeger & Heinzl, 2021). Collectively, they demonstrate the strong negative impact of breakdowns and underscore the need to address breakdowns when they occur.

Third, repair strategy exploration studies describe the various strategies available for addressing breakdowns in customer-chatbot interaction. These studies typically extract repair strategies from the literature or by experimenting with real-world chatbots (e.g., Benner et al., 2021; Feng, 2023). The identified strategies can be distinguished into two main types: customer-driven and chatbot-driven (Alghamdi et al., 2024). In customer-driven strategies, the chatbot typically responds to a breakdown by asking the customer to repeat or reformulate their previous input (e.g., "Sorry, I didn't understand that. Please try again"; Feng, 2023). In contrast, chatbot-driven strategies tend to have the chatbot try to guess what the customer is asking and provide the response that is most likely to be appropriate, usually without acknowledging that a breakdown occurred in the first place (Benner et al., 2021). Although most of these studies are conceptual in nature, some also explore customers' preferences for different repair strategies, indicating a tendency toward strategies that do not leave it entirely up to the customer to resolve a breakdown (Ashktorab et al., 2019). Overall, these studies provide a good overview of the current landscape of repair strategies and the complexities associated with implementing them.

Finally, repair strategy impact studies empirically investigate how different repair strategies affect customer perceptions and attitudes toward a chatbot and the organization behind it. While most of these studies select one repair strategy and test it against a control condition without that strategy (e.g., Han et al., 2021; Sheehan et al., 2020; Weiler et al., 2022), some also evaluate multiple strategies (e.g., Braggaar et al., 2023; Yeh et al., 2022). Overall, the results of these studies indicate positive effects of chatbot-driven repair strategies, such as providing options or asking for clarification, on perceptions of the chatbot (e.g., competence, warmth), customers' emotions (e.g., anger, aggression), and downstream outcomes (e.g., satisfaction, service quality, reuse intention) (e.g., Han et al., 2022; Haupt et al., 2023; Huang & Dootson, 2022). Although these studies rely primarily on controlled laboratory settings rather than real-world interactions, they suggest that employing repair strategies is a viable approach to mitigating the negative impacts of breakdowns.

Despite the considerable amount of existing research, three important knowledge gaps remain. First, prior research has largely focused on repair strategies that tend to put the onus of resolving breakdowns on either the customer or the chatbot alone. One exception is the scenario-based study by Ashktorab et al. (2019), which revealed an overall preference for more collaborative repair strategies (e.g., having the chatbot provide suggestions, ask for confirmation, or offer explanations) but did not examine these strategies and their success rates in real chatbot interactions. Consequently, although there is some indication that customers are willing to resolve breakdowns together with a chatbot, we know little about how to design such collaborative repair strategies and whether they actually help overcome breakdowns in customer-chatbot interaction. Second, there is a lack of research on the impact of (collaborative) repair strategies in real-world settings where breakdowns occur naturally. Although existing studies offer valuable insights into customers' evaluations of various repair strategies (e.g., Han et al., 2022; Haupt et al., 2023; Weiler et al., 2022), controlled settings may fail to reflect customers' real experiences with chatbots and cannot account for the complexity of actually implementing repair strategies (Ashktorab et al., 2019; Law et al., 2022). This is an important gap because without capturing customer behavior under real-world conditions, it is difficult to assess a repair strategy's effectiveness. Finally, existing empirical studies have adopted a unidimensional view of breakdowns without considering potential differences in their underlying causes (Feng, 2023; Rapp et al., 2023). As these differences might influence the effectiveness of repair strategies, a more nuanced picture of breakdowns is needed. Following the DSR paradigm, we addressed these gaps by proposing a theory-driven design for collaborative repair strategies and investigating its impact in real-world customer-chatbot interaction.

A Theory-Driven Design for Collaborative Repair Strategies

This research follows the DSR paradigm (Hevner et al., 2004). DSR aims to provide general design solutions for a class of real-world problems rather than a single instance (Gregor & Hevner, 2013). Our research addresses the class of problems that we refer to as conversational breakdowns in customer-chatbot interaction. Given their negative impacts and the difficulty of avoiding such breakdowns altogether, there is a need for effective repair strategies; existing strategies tend to result in frustrated and dissatisfied customers. Since the typical repair strategy tends to put the onus of resolving breakdowns on either the customer or the chatbot alone (see Appendix A), we propose designing collaborative strategies inspired by how humans work together to resolve breakdowns in everyday conversation.

However, it is unclear how this fundamental characteristic of human communication can be translated into a form that can guide the design of collaborative repair strategies in real-world customer-chatbot interaction. We therefore followed the DSR approach described by Walls et al. (1992) to develop a theory-driven design for collaborative repair strategies consisting of an integrated kernel theory, meta-requirements, a meta-design, and testable propositions. Figure 1 illustrates our proposed design. Like prior DSR studies (e.g., Li et al., 2020b), we discuss the details of each element in the following subsections.

Kernel Theory

A fundamental component of DSR is a solid theoretical grounding that justifies the proposed design and explains why it should work to address the design problem at hand (Gregor & Hevner, 2013). To this end, DSR researchers typically select a kernel theory from the natural or social sciences to govern the design requirements (Walls et al., 1992). Although such theories originating from disciplines outside IS can rarely be used "as is" to guide the design of an IT artifact, they should nevertheless come from domains closely associated with the problem domain of the artifact (Kuechler & Vaishnavi, 2008, 2012). As our research is concerned with communication problems and how they can be resolved, we selected a kernel theory from the human-human communication literature and then integrated the perspective of customer-chatbot interaction.

Human communication theories provide an appropriate theoretical lens for situations in which one communication partner is not human, as people instinctively draw on their knowledge of human communication to make sense of and guide their interactions with technology (Guzman & Lewis, 2020; Westerman et al., 2020). Under the "computers are social actors" paradigm, many studies have shown that people respond to technology as they would to another person, even though they know that their interlocutor is not human (Reeves & Nass, 1996). This has been observed across various technologies, including customer service chatbots (e.g., Chandra et al., 2022; Schanke et al., 2021). Hence, communication researchers maintain that there is no conceptual necessity to restrict their theories to interactions between humans (Guzman, 2018; Rosenthal-von der Pütten & Koban, 2023).

Against this backdrop, we selected the theory of least collaborative effort (TLCE; Clark & Wilkes-Gibbs, 1986) as our theoretical lens. This theory is well-suited for our phenomenon of interest because it explains how humans deal with breakdowns in everyday conversation, has been supported by evidence across various languages and communication media (Dingemanse et al., 2015), and has been used in prior studies on chatbots (e.g., Dippold, 2023). While other theories take a broader look at human communication (e.g., Habermas's theory of communicative action), TLCE provides a detailed account of the fundamental principles governing the process by which humans identify and resolve communication breakdowns. It is therefore a good fit for informing the design of repair strategies in customer-chatbot interaction.

Although TLCE provides a solid theoretical grounding to inform our design, its use as a kernel theory requires an additional step, commonly referred to as the "conceptual leap" in DSR (Kuechler & Vaishnavi, 2008). Due to differences in scope, granularity, and boundary conditions, kernel theories typically need to be adapted to the specific IS phenomenon before they can adequately guide the design of an IT artifact (Arazy et al., 2010). Therefore, we integrated TLCE with findings from the bodies of literature on human-machine communication and customer service chatbots. In the following, we first introduce the original TLCE and then present the framework resulting from our integration (TLCE-bot), which served as the kernel theory guiding our design of collaborative repair strategies.

Theory of Least Collaborative Effort (TLCE)

According to TLCE, humans try to minimize the collaborative effort required to reach mutual understanding in communication (Clark & Wilkes-Gibbs, 1986). Departing from classical theories of least effort that imply egoistic motivations behind communication behavior (e.g., Zipf, 1949), TLCE asserts that human communication is a fundamentally collaborative activity: Humans try to minimize not just their own effort but rather the joint effort for themselves and their counterparts (Clark & Brennan, 1991). Among the communication phenomena that TLCE accounts for is the process of conversational repair, through which humans address problems of speaking, hearing, and understanding in communication (Schegloff et al., 1977). This process generally consists of a three-part sequence: a trouble source (such as an unfamiliar word), followed by a repair initiation (signaling a problem with the trouble source) and a candidate repair solution (the actual repair, such as paraphrasing the unfamiliar word). On average, such sequences occur every 84 seconds in human communication (Enfield, 2017). There is a central distinction between who initiates the repair (self or other) and who performs it (self or other), where the self is the speaker of the trouble source and the other is its recipient in the conversation. TLCE describes three principles governing the collaborative process of conversational repair, referred to here as the principles of (1) self-repair preference, (2) specificity, and (3) incremental upgrading.

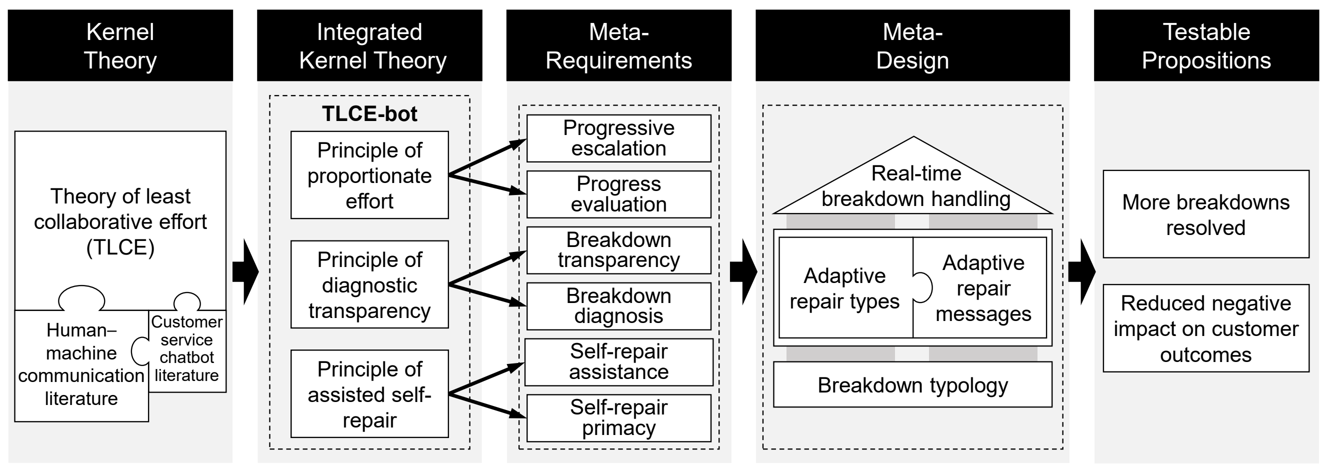

Figure 1. Theory-Driven Design for Collaborative Repair Strategies

First, there is a fundamental preference for self-repair over other-repair in human communication (Schegloff et al., 1977), which we refer to as the *principle of self-repair preference*. Humans prefer to repair their own trouble sources and therefore tend to allow speakers to do the same rather than explicitly correcting them. This is because other-repair can be a sensitive issue and may even be interpreted as rude (Hutchby & Wooffitt, 1998). Moreover, performing the repair usually requires less effort for the speaker than for the recipient because the speaker can more quickly identify an adequate repair solution (Clark & Brennan, 1991). Second, when humans initiate repair, they choose among classes of repair initiators that vary in specificity and thereby power in terms of locating and repairing the trouble source (Dingemanse et al., 2015; Schegloff et al., 1977). For example, open-class repair initiators (e.g., "Huh?") contain little information about the trouble source, leaving it entirely up to the speaker of the trouble source to resolve the breakdown. To minimize collaborative effort, recipients tend to choose the most specific repair initiator possible—a phenomenon referred to as the *principle of specificity* or the strongest initiator rule (Clark & Schaefer, 1987; Dingemanse et al., 2015). For example, a more specific repair initiator might be a restricted request (e.g., "Who?" or "Where?") that makes the trouble source explicit and typically requires a shorter repair solution. Third, while the repair process would ideally yield a resolution after the first repair attempt, it is not uncommon for multiple attempts to be necessary (Schegloff et al., 1977). In such situations, humans tend to "upgrade" to incrementally more effortful or complicated types of repair until the breakdown is resolved (Albert & de Ruiter, 2018; Svennevig, 2008). We refer to this as the *principle of incremental upgrading*. To minimize collaborative effort, humans first try less effortful repair types (e.g., self-repair by asking for clarification) before moving on to more effortful types of repair (e.g., other-repair by proposing a candidate repair solution) (Svennevig, 2008).

Theory of Least Collaborative Effort in Customer-Chatbot Interaction (TLCE-bot)

TLCE provides valuable insights into the fundamental principles of how humans identify and resolve breakdowns in communication. Nevertheless, its focus on human communication partners raises questions about its applicability to our context, in which the communication is between a human (customer) and a machine (chatbot). To bridge the incongruity between TLCE and our context, we followed Jiang et al.'s (2022) conceptual blending approach to integrate TLCE with findings from the bodies of literature on human-machine communication and customer service chatbots. Conceptual blending involves taking the elements of two or more input mental spaces and combining them in creative ways to draw novel insights about a phenomenon of interest (Oswick et al., 2011). First, the human-machine communication literature provides an appropriate mental space due to its long history of exploring how and why human interactions with machines resemble or differ from human-human communication. On the one hand, studies in this area have shown that social scripts and rules of human communication, such as politeness and small talk, are also used in chatbot interactions (Dippold, 2023; Nguyen et al., 2021). On the other hand, this literature also highlights that, as long as they are distinguishable, humans do not treat chatbots identically to human beings in all aspects of communication (Brendel et al., 2023; Guzman & Lewis, 2020). Second, prior literature on customer service chatbots offers valuable complementary insights into the specific domain of online customer service, where customers communicate with chatbots to quickly get answers to their questions, obtain product information, and resolve issues (e.g., Chandra et al., 2022; Crolic et al., 2022; Meier et al., 2024). Drawing on findings from these two bodies of literature, we used an interrogative technique, as suggested by Jiang et al. (2022), to observe and challenge the implicit and explicit assumptions of TLCE's three principles. This approach facilitated the identification of both conceptual overlaps and structural differences between the domains, enabling us to develop new principles that explicate the dynamics of breakdown and repair in the domain of customer-chatbot interaction. Similar approaches have been used in prior DSR studies, where the result is sometimes called "design relevant explanatory/ predictive theory" (Kuechler & Vaishnavi, 2008) or "applied behavioral theory" (Arazy et al., 2010). We refer to the framework resulting from our integration as *TLCE-bot* and elaborate on its three principles below.

First, TLCE's principle of self-repair preference states that humans prefer to repair their own trouble sources rather than being explicitly corrected by others, and they therefore tend to provide opportunities for others to repair their trouble sources as well. As a result, the most common repair type is other-initiated self-repair, where the recipient (other) indicates a problem but then allows the speaker (self) to repair the trouble source (Schegloff et al., 1977). Conceptual blending of this principle with findings from the literature on human-machine communication and customer service chatbots yielded a mixed picture. Consistent with the principle, prior research has shown that customers will themselves try to resolve breakdowns in their chatbot interactions (Rapp et al., 2023), highlighting that other-initiated self-repair is also the prevalent type of repair in human-machine communication, with the chatbot (other) signaling a breakdown and the customer (self) resolving it (Benner et al., 2021). However, the literature has also found that customers often fail to resolve breakdowns on their own (Dippold, 2023; Li et al., 2020a) and may even prefer other-repair by a human employee for complex problems (Braggaar et al., 2023). Therefore, departing from the original principle of self-repair preference, some studies on customer service chatbots have suggested that customers may require assistance for their self-repair attempts, rather than just an opportunity to resolve the breakdown on their own (Ashktorab et al., 2019; Haupt et al., 2023). For example, a chatbot could offer guidance or tips on how to reformulate a request for better understanding (Nguyen et al., 2021; Yeh et al., 2022). Customers may then welcome explicit assistance in resolving breakdowns together with the chatbot—something that would be considered unnatural and unnecessary in human communication. Taken together, our findings suggest that while the prevalence of self-repair over other-repair is similar in customer-chatbot interaction, customers count on assistance from the chatbot rather than being left to guess how the breakdown can be resolved. Therefore, we propose the *principle of assisted self-repair*: Customers need and appreciate assistance from the chatbot in self-repairing breakdowns.

Second, TLCE's principle of specificity states that when humans initiate the repair process, they tend to choose the most specific repair initiator possible to help the speaker locate the source of trouble (e.g., by asking "Who?" or "Where?" instead of "Huh?"). Conceptual blending of this principle with findings from the literature on human-machine communication and customer service chatbots revealed that chatbots do not adhere to this principle. Instead, they often initiate repair using a generic message that does not help customers understand the source of the breakdown (e.g., "Sorry, I didn't understand that. Please try again.") (Dippold, 2023; Li et al., 2020a). This disparity may be due to the difficulty of adapting the principle of specificity to the domain of customer-chatbot interaction. On the one hand, chatbots may not be able to fully understand the nuances of human language necessary to precisely locate the breakdown's source and respond with a more specific repair initiator (Kopp & Krämer, 2021). On the other hand, prior studies have shown that humans in general, and customers in particular, may communicate with chatbots in a way that makes it difficult to establish exactly where the problem lies. For example, customers often submit requests that provide irrelevant details or not enough information (e.g., only a keyword) (Li et al., 2020a; Verne et al., 2022). This may be due to a lack of awareness on how to effectively communicate with a chatbot (Dippold, 2023) or a tendency to invest less effort in formulating requests when communicating with a chatbot compared to another human (Corti & Gillespie, 2016). As customer service chatbots typically take the place of human service employees, customers tend to expect them to demonstrate the same interactional competencies that are associated with humans (Chandra et al., 2022). Against this backdrop, we suggest a focus on the logic underlying the principle of specificity: facilitate the understanding of what is causing the problem in communication. Even if the source of a breakdown cannot be pinpointed, any information that helps customers understand its cause may also enable them to resolve it (Feng, 2023; Yeh et al., 2022). Furthermore, given the wide variety of causes that can lead to a breakdown (Feng, 2023; Rapp et al., 2023), such transparency may not only allow customers to direct their repair efforts more effectively but also educate them about how to better communicate with the chatbot in general (Dippold, 2023). Therefore, we propose the *principle of diagnostic transparency*: Breakdowns are more likely to be resolved when the chatbot provides customers with information about the cause of a breakdown.

Third, TLCE's principle of incremental upgrading states that when the repair process fails to yield a resolution after the first attempt, humans upgrade to incrementally more effortful or complicated types of repair until the trouble source is resolved. Conceptual blending of this principle with findings from the literature on human-machine communication and customer service chatbots revealed that upgrades after consecutive breakdowns in customer-chatbot interaction are quite different in nature. Upgrades on the human side of the interaction typically involve rephrasing a request (e.g., omitting certain details or using synonyms) (Li et al., 2020a; Rapp et al., 2023). In contrast, some chatbots exhibit no upgrading at all when consecutive breakdowns occur, leaving customers trapped in loops of trial and error with the same default response from the chatbot (Nguyen et al., 2021; Rapp et al., 2023). Other chatbots proceed immediately to more effortful repair types, such as redirecting customers to a website or call center, rather than helping them resolve the breakdown within the ongoing interaction (Nguyen et al., 2021; Verne et al., 2022). Although some organizations employ human employees to step in and resolve breakdowns, this approach not only faces scalability issues but also creates additional costs (Braggaar et al., 2023; Crolic et al., 2022; Huang & Dootson, 2022). Taken together, this illustrates the dilemma in customer-chatbot interaction: Whereas chatbots never lose patience and are always willing to answer questions (Meier et al., 2024), customers become impatient and eventually give up on the chatbot if there is no progress in the conversation (Li et al., 2020a). At the same time, customers may or may not like being redirected, but certain requests indeed require the involvement of a higher authority because they cannot, or should not, be handled by a chatbot (Huang & Dootson, 2022). This highlights the importance of striking a balance between the severity of a breakdown in customer-chatbot interaction and the level of effort required to resolve it. Just as it may be inappropriate for a chatbot to immediately redirect customers instead of giving them an opportunity for self-repair, it may also be unproductive to repeatedly prompt a customer to try again without providing further assistance (e.g., additional tips or instructions). Therefore, we propose the *principle of proportionate effort*: The level of effort required from both the customer and the chatbot in resolving a breakdown corresponds to the severity of the breakdown situation.

Meta-Requirements

Based on TLCE-bot, we derived a set of meta-requirements for the design of collaborative repair strategies in customer-chatbot interaction. To this end, we translated the three principles of TLCE-bot from the theoretical realm to the design space, resulting in two requirements for each principle (Kuechler & Vaishnavi, 2012).

According to the principle of assisted self-repair, customers need and appreciate assistance from the chatbot in self-repairing breakdowns. Following this principle, collaborative repair strategies should meet two requirements. First, consistent with the self-service nature of customer service chatbots, they should focus on providing opportunities for customers to resolve breakdowns themselves (*self-repair primacy requirement*). Second, they should offer assistance to customers throughout the repair process (*self-repair assistance requirement*). Without assistance, customers have to guess what the chatbot can understand, leaving them to either resort to a trial-and-error approach or abandon the chatbot immediately.

The principle of diagnostic transparency states that breakdowns are more likely to be resolved when the chatbot provides customers with information about the cause of a breakdown. Two requirements follow from this principle. First, collaborative repair strategies should be able to diagnose the most likely cause of a breakdown in real time (*breakdown diagnosis requirement*). Despite the difficulty of pinpointing the exact cause of a breakdown in customer-chatbot interaction, any information that helps customers understand what went wrong may enhance their ability to resolve the breakdown together with the chatbot. Therefore, the second requirement is to include information that is as specific as possible about the cause of a breakdown to help customers understand it (*breakdown transparency requirement*). Without transparency, customers will struggle to effectively direct their repair efforts.

According to the principle of proportionate effort, the level of effort required from both customers and the chatbot in resolving a breakdown corresponds to the severity of the breakdown situation. For example, in situations where consecutive breakdowns occur, it is likely that both the customer and the chatbot must invest extra effort to achieve a resolution. From this principle, we also derived two requirements. First, collaborative repair strategies should be able to monitor and acknowledge the progress, or lack thereof, in resolving the breakdown over the course of the interaction (*progress evaluation requirement*). This not only informs customers about the progress in their interaction with the chatbot but also highlights the chatbot's ability to evaluate the severity of the breakdown situation. Using the monitoring results as a basis, collaborative repair strategies should then progressively increase the level of assistance until the breakdown is resolved or, as a last resort, the customer is redirected to a human employee (*progressive escalation requirement*). Without progressive escalation, customers will find themselves in trial-and-error loops, lose patience, and ultimately leave in frustration.

Meta-Design

To meet the identified requirements, we propose a meta-design for collaborative repair strategies that consists of four interrelated components: (1) breakdown typology, (2) adaptive repair types, (3) adaptive repair messages, and (4) real-time breakdown handling. Following the DSR paradigm, our meta-design describes a class of solutions rather than a particular instance of one solution (Walls et al., 1992). In the following subsections, we introduce each of the four components before explaining how they were instantiated at InsurCo in the subsequent section.

Breakdown Typology

Just as breakdowns in human communication can arise from a range of trouble sources, breakdowns in customer-chatbot interaction may also stem from a variety of causes. Without a clear understanding of these causes, it would be difficult to diagnose which one is most likely behind a specific breakdown (breakdown diagnosis requirement) and provide customers with information about the breakdown (breakdown transparency requirement), as well as assistance in resolving it (self-repair assistance requirement). Hence, the first component of a collaborative repair strategy is a chatbot-specific breakdown typology that categorizes breakdowns according to common characteristics and causes. An ideal starting point for developing the typology would be to examine conversation logs, breakdown-causing messages, and other data from customer interactions with the chatbot that is to be equipped with the new repair strategy. This bottom-up approach will not only provide a detailed view regarding when and why breakdowns occur but will also allow for the identification of information (e.g., characteristics of breakdown messages) that can be used to classify breakdowns into distinct types. While such a classification could be performed manually, unsupervised machine learning approaches (e.g., cluster analysis) are well-suited for identifying patterns and grouping similar breakdown messages into a set of breakdown types. These breakdown types can then serve as the foundation for the collaborative repair strategy and provide the basis for all subsequent components.

Adaptive Repair Types

Building on the chatbot-specific breakdown typology, the second component of a collaborative repair strategy consists of adaptive repair types, which the chatbot can use to initiate repair after a breakdown. These go hand-in-hand with adaptive repair messages (see next subsection) because certain repair types require certain repair messages, and vice versa. Based on the idea of adaptive user interfaces (Reinecke & Bernstein, 2013), the adaptation logic is responsible for automatically providing a repair type that is adapted to the individual situation rather than using the same type of repair whenever a breakdown occurs. Following the self-repair primacy requirement, the focus here is on customer self-repair. However, if customers are unlikely to be able to resolve breakdowns themselves, the chatbot may also select a type of other-repair (e.g., redirecting the customer to a human-operated service channel) instead of self-repair. Importantly, in line with the progress evaluation requirement, this component takes into account both the current breakdown type and those of any previous breakdowns. As a result, when multiple breakdowns occur, repair types can be adapted to each consecutive breakdown. Rather than following a one-size-fits-all approach, the chatbot can move from less to more effortful repair types in the repair process, thereby addressing the progressive escalation requirement. Typically, the least effortful repair types focus on customer self-repair, while the most effortful ones include other-repair by human service employees who resolve a breakdown outside the ongoing customer-chatbot interaction. Repair types involving human employees are therefore often a last resort after all other repair types have failed (Benner et al., 2021). Prior to human involvement, there should be an increase in the level of assistance provided by the chatbot, such as extending the specific guidance in its repair message with some general rule-based guidance (Yeh et al., 2022). Ultimately, the instantiation of the adaptive repair type component should result in a set of carefully selected repair types for all possible breakdown situations.

Adaptive Repair Messages

Closely intertwined with the adaptive repair type component and also building on the breakdown typology is the third component of a collaborative repair strategy: adaptive repair messages. In line with the idea of adaptive user interfaces (Reinecke & Bernstein, 2013), these repair messages adapt to individual breakdown types instead of sending the same text whenever a breakdown occurs. Their main purpose is to help customers identify the problem with their input and thus enable them to resolve the breakdown themselves, thereby addressing the breakdown transparency and self-repair assistance requirements. In addition, these messages serve to acknowledge the progress made in resolving a breakdown, thereby fulfilling the progress evaluation requirement.

To support the implementation of this component, our meta-design includes a conceptual schema for adaptive repair messages comprising three elements: apology, explanation, and guidance. Based on this schema, generic repair messages (e.g., “Sorry, I didn’t understand that. Please try again.”) can be replaced with specific ones adapted to individual breakdown types. As the examples in Table 1 illustrate, the goal is to offer breakdown type-specific guidance that puts customers in a position to resolve breakdowns themselves. To understand the kind of guidance that customers need to perform a successful self-repair, it is important to determine the underlying cause of each breakdown type identified and to examine breakdown-causing messages. Our schema also includes the elements of apology and explanation. While an apology acknowledges the chatbot’s responsibility, the explanation element offers breakdown type-specific insights that help customers identify the problem with their input. Overall, the instantiation of the adaptive repair message component should result in a set of repair messages with specific explanations and guidance adapted to individual breakdown types.

Table 1. Conceptual Schema for Adaptive Repair Messages

Element	Description	Example
Apology	Expresses regret and accepts responsibility for the breakdown	*“I’m sorry.”*
Explanation	Provides insights into what has caused the breakdown	*“I sometimes struggle with long requests.”*
Guidance	Offers specific instructions on how to resolve the breakdown	*“Can you please repeat your request with fewer details?”*

Real-Time Breakdown Handling

The repair strategy components just presented can only realize their potential if the chatbot is able to analyze breakdown situations in real time and then initiate the repair process in an adaptive manner. This task is carried out by the real-time breakdown-handling component: Whenever a breakdown occurs, this component classifies the customer’s input into one of the breakdown types (breakdown diagnosis requirement), evaluates any previous customer self-repair attempts (progress evaluation requirement), and selects a repair type and repair message adapted to the individual situation (progressive escalation requirement). The real-time classification can be done through rule-based approaches, machine learning approaches, or a combination of the two. For example, a supervised machine learning approach could be used to train a breakdown type classifier using real-world customer messages as training data. In this case, it is important to focus on data available to the chatbot in its runtime environment; otherwise, the chatbot will be unable to classify breakdown types in real time when the repair strategy is implemented. The final step is to integrate this component into the running chatbot so that it can take over whenever breakdowns occur.

Testable Propositions

Evaluation in DSR is commonly guided by testable propositions regarding the extent to which the proposed design satisfies the meta-requirements (Walls et al., 1992). In the present case, it is important to evaluate whether the proposed design of collaborative repair strategies can (1) help customers and chatbots resolve breakdowns and (2) mitigate the negative impacts of breakdowns on key customer outcomes (e.g., satisfaction). To do this, a multifaceted evaluation with a comparison against a baseline repair strategy is needed. Specifically, the first proposition requires an empirical investigation of customer-chatbot interaction and an analysis of objective data regarding whether breakdowns were resolved. The second calls for subjective customer evaluations of breakdown situations and the repair strategy implemented in the chatbot. Both facets of the evaluation should reveal a collaborative repair strategy to be more effective than a baseline strategy that does not address our requirements.

Design Instantiation and Evaluation at InsurCo

To instantiate, implement, and evaluate our theory-driven design for collaborative repair strategies under real-world conditions, we collaborated with InsurCo, a large international insurance company. Their AI-based customer service chatbot, which is available on web pages for household, term life, personal liability, and pet health insurance, uses natural language processing techniques to answer common questions in German, with more than 300 answers available for each insurance type. Although these answers were developed iteratively and trained continuously over several months, breakdowns in customer interactions with InsurCo’s chatbot seemed inevitable. Therefore, when we approached InsurCo’s managers with our design for a collaborative repair strategy, they showed great interest and agreed to participate in our research project. It should be noted that while we appreciated their feedback on our ideas, the intellectual work and the development of the artifacts were solely our responsibility. Given that their chatbot used a relatively basic customer-driven repair strategy at the time, InsurCo presented an ideal setting for instantiating our proposed design and evaluating its impact on real-world customer-chatbot interaction.

The instantiation of our proposed design for collaborative repair strategies followed the notion of a “design journey” (vom Brocke et al., 2020) in which problem space, solution space, and evaluation activities were combined to accumulate design knowledge over time. Often, DSR projects are structured along iterations of the DSR process, where a single artifact is developed in a series of design cycles (Kuechler & Vaishnavi, 2008). However, rather than concentrating on a single artifact, we focused on the four components of our proposed meta-design—breakdown typology, adaptive repair types, adaptive repair messages, and real-time breakdown handling—which we successively instantiated, evaluated, and eventually implemented in InsurCo’s chatbot. To reflect this approach, we use the term “design episode” to refer to “moments when new IT capabilities were added, modified, expanded, or purged” (Hanseth & Lyytinen, 2010, p. 8). After the initial problem space exploration phase (for details, see Reinkemeier & Gnewuch, 2022), we structured our DSR project around three design episodes and concluded with a summative evaluation in a naturalistic setting. Figure 2 provides an overview of our DSR journey. In the following subsections, we detail the activities and results of the three episodes and the final evaluation.

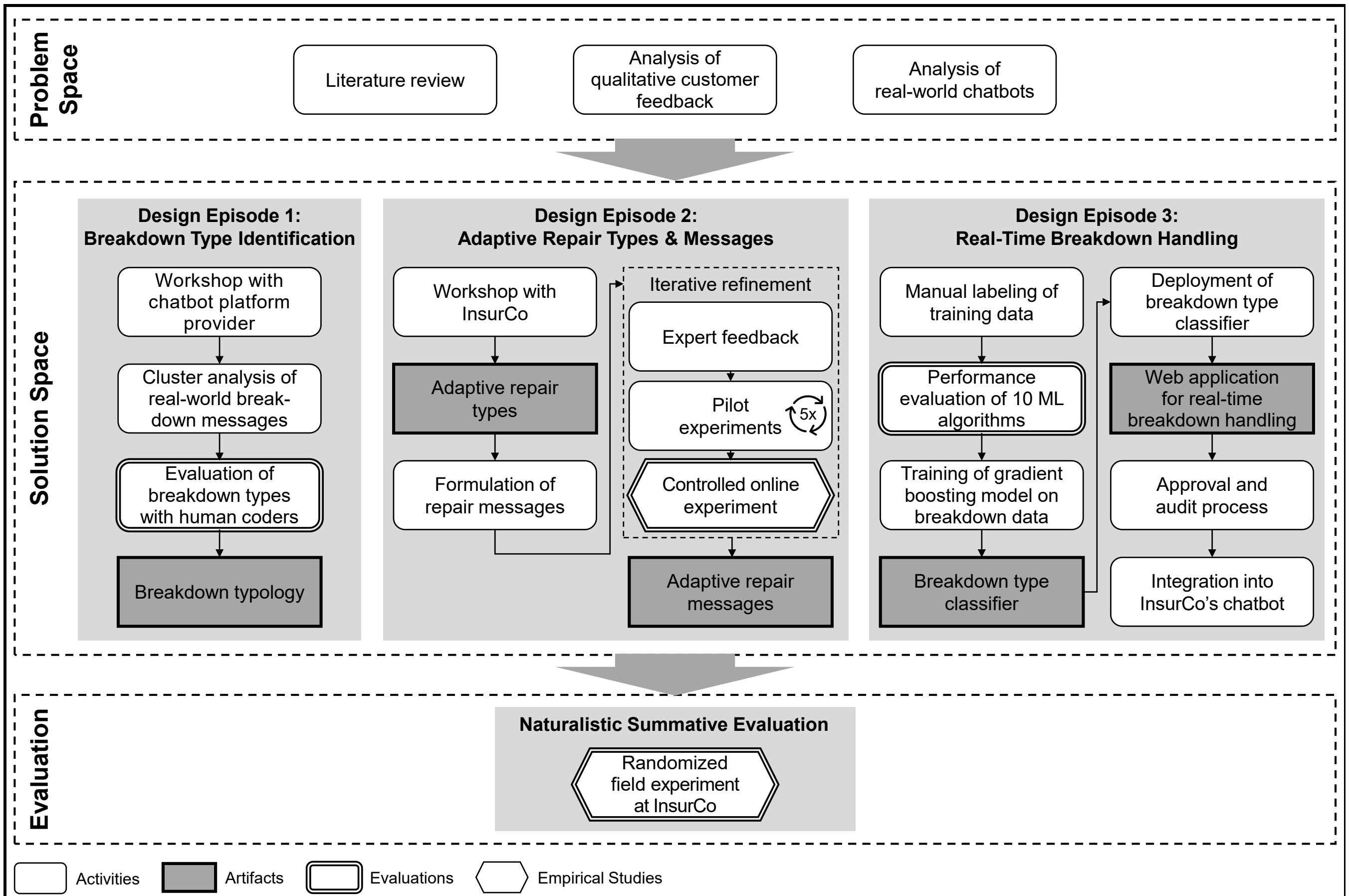

Figure 2. Design Science Research Journey

Design Episode 1: Breakdown Type Identification

Our first design episode focused on instantiating the fundamental component of a collaborative repair strategy, namely, a chatbot-specific breakdown typology. To identify the different types of breakdowns in customer interactions with InsurCo's chatbot, we leveraged a dataset of 21,736 real-world customer messages and performed a cluster analysis on the 5,668 messages (26%) that had caused a breakdown (hereafter referred to as breakdown messages). To uncover more generalizable breakdown causes, our analysis focused on four content-independent characteristics of breakdown messages: semantic weight, percentage of unknown words, word count, and words per sentence. Semantic weight was an indicator of the wealth of information in a message, while the unknown words were words that did not exist in the chatbot's knowledge base. Our cluster analysis resulted in four clusters (for details, see Reinkemeier & Gnewuch, 2022), each representing a distinct breakdown type: (1) elaborate, (2) unique, (3) brief, and (4) cryptic (see Table 2).

Finally, we evaluated our typology with a separate dataset by instructing two human coders to manually assign 180 breakdown messages to one of the four breakdown types. The agreement between their assignment and an automated classification based on our typology was 79.44%, with a corresponding Cohen's kappa of 0.70 (substantial agreement; cf. Landis & Koch, 1977), which indicates the validity of the breakdown types as a solid starting point for the next design episodes.

Design Episode 2: Adaptive Repair Types and Adaptive Repair Messages

Building on the four breakdown types identified in the first episode, our second episode focused on instantiating the two closely interrelated components of adaptive repair types and adaptive repair messages. We first conducted a workshop with InsurCo to discuss our breakdown types and determine which repair type to use in which situation. We then formulated and refined repair messages for each breakdown type and evaluated them in a series of controlled online experiments.

Table 2. Breakdown Typology Based on Cluster Analysis Results

Breakdown type		Number of messages	Message characteristic (mean)				Example messages
			SW	UW	WC	WPS	
1	**Elaborate**	763 (13.46%)	7.89	4.62	21.75	14.95	"hello I would like to take out a household insurance with you but do not know exactly the square meter specification of the apartment. is the approximate specification also in order"
2	**Unique**	1,872 (33.03%)	4.32	5.27	10.96	9.63	"Isn't the bicycle as a sprts [*sic*] equipment already insured worldwide anyway?"
3	**Brief**	2,516 (44.39%)	1.80	5.40	4.20	4.07	"Are wallpapers also insured", "fear"
4	**Cryptic**	517 (9.12%)	0.88	83.22	1.97	1.94	"tililes", "Vorrei sapere come fare l assicurazione casa a berlino"

Note: SW = semantic weight; UW = percentage of unknown words; WC = word count; WPS = words per sentence. Original spelling and punctuation preserved for all examples (translated from German).

Determination of Repair Types

Based on our discussions with InsurCo, we decided to address the first breakdown of any type with "assisted customer self-repair," in which the chatbot provides assistance in the form of guidance and explanations to enable customers to self-repair the breakdown. Although most failed interactions included only one breakdown (after which customers usually abandoned the chatbot), there were some with two or, in rare cases, three consecutive breakdowns. We decided that after three breakdowns, customers should always be offered the opportunity to get in touch with a human employee (e.g., via a callback option), as it would be unlikely at this point that the breakdown could be resolved without external support. For situations involving two consecutive breakdowns, we decided to use different repair types for different combinations of breakdown types. For example, after two consecutive brief-type breakdowns, customers would receive another assisted self-repair opportunity, but this time extended with additional rule-based guidance (e.g., "It often helps to use alternative words with similar meanings"). Ultimately, our discussions resulted in a well-defined set of repair types that served as our foundation for the following steps.

Formulation and Refinement of Repair Messages

The second step of Design Episode 2 consisted of formulating repair messages based on our proposed schema for adaptive repair messages (see Table 1). In this step, we specifically focused on the guidance and explanation elements of the repair messages used for assisted customer self-repair. After scrutinizing each breakdown type and examining typical breakdown messages, we created an initial set of 25 repair messages. For example, the repair message for brief-type breakdowns was initially formulated as "I'm sorry! I struggle with short requests. Please repeat your request with additional information." To evaluate whether these repair messages would assist customers in self-repairing breakdowns, we conducted a series of controlled online experiments in which we compared our breakdown type-specific repair messages (see Table 3 for final versions) against a generic version ("I'm sorry! I didn't understand that. Please try again."). All experiments employed a 2 (repair message: specific vs. generic) × 4 (breakdown type: 1-4) between-subjects design. In each experiment, participants were asked to imagine themselves in a hypothetical but realistic scenario related to the coverage of homeowners insurance.

To create a realistic setting with high internal validity, we developed a custom chatbot and prepared eight conversations with a predefined breakdown message from the customer (Type 1-4) and a repair message from the chatbot (specific or generic). This enabled us to randomly assign participants to one of the eight conditions and allowed participants to begin interacting with the chatbot after a breakdown had already occurred. Our custom chatbot was inspired by InsurCo's chatbot and used the same natural language processing techniques. As Figure 3 illustrates, the predefined conversations were structured into three parts: (1) chatbot introduction, (2) breakdown message, and (3) repair message. Participants were tasked with reading the conversation and following the chatbot's instructions. After participants entered one message (their repair attempt), the conversation ended. They received no further response from the chatbot so as not to influence their evaluation of the chatbot and to keep the conditions comparable.

Table 3. Overview of Chatbot Repair Messages Adapted to Breakdown Types (Translated From German)

Breakdown type	Repair message		
	Apology	Explanation	Guidance
1 (Elaborate)	I'm sorry!	I sometimes struggle with long requests.	Therefore, I'd like to ask you to shorten your request to the most important information.
2 (Unique)	I'm sorry!	I sometimes struggle with certain formulations.	Therefore, I'd like to ask you to repeat your request using different words.
3 (Brief)	I'm sorry!	I sometimes struggle with short requests.	Therefore, I'd like to ask you to repeat your request with additional important information.
4 (Cryptic)	I'm sorry!	I am not sure whether your request is within the scope of my knowledge.	Therefore, I'd like to ask you to repeat your request as a detailed question with all the important information.

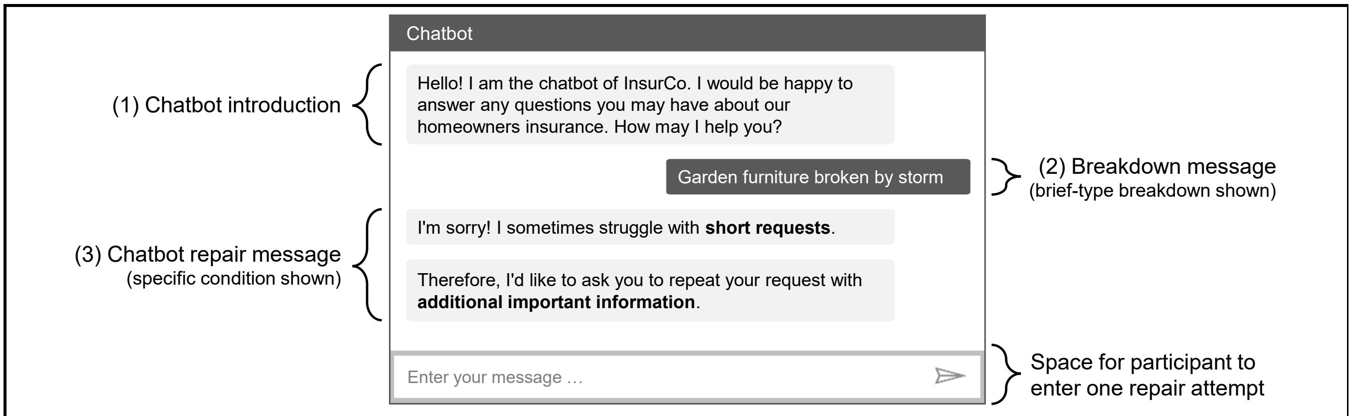

Figure 3. Screenshot of the Chatbot Interface in the Online Experiment (Translated From German)

After extensive pretesting and iterative refinements based on five pilot experiments, we were confident about our set of repair messages (see Table 3) and proceeded with our main experiment. Of the 591 participants who completed the experiment, we excluded 12 because they did not follow the instructions. Therefore, our final sample consisted of 579 participants (41.45% female, average age = 38 years), almost equally distributed among the eight conditions (69-75 participants per condition). Randomization checks revealed no systematic differences between the conditions based on age, gender, or chatbot experience. In addition, our manipulation check showed that participants who received a specific repair message ($M = 4.39$, $SD = 1.48$) perceived that the chatbot had assisted in solving communication problems to a greater extent than did those who received the generic message ($M = 3.34$, $SD = 1.45$; $t(577) = 8.62$, $p < 0.001$), confirming that our manipulation was successful.

To assess the impact of specific (vs. generic) repair messages for each breakdown type, we analyzed (1) the scores for the chatbot's ability to correctly recognize the intent in the messages entered by the participants (i.e., their repair attempt) and (2) the participants' subjective evaluation of the chatbot. Given that the intent recognition score indicates how well the chatbot understands the customer input, it is a viable proxy for how successful participants were in acting upon the chatbot's guidance and self-repairing the breakdown. First, we examined the results for all breakdown types taken together using the non-parametric Mann-Whitney U test and found that specific repair messages led to significantly higher intent recognition scores ($M = 0.611$) than generic messages ($M = 0.499$, $z = 6.570$, $p < 0.001$, $r = 0.273$). We then used follow-up Mann-Whitney U tests to conduct Bonferroni-adjusted pairwise comparisons between the specific and generic conditions for each of the four breakdown types. As shown in Table 4, the intent recognition scores were significantly higher in the specific condition than in the generic condition for elaborate-, unique-, and brief-type breakdowns (all p values < 0.01). For cryptic-type breakdowns, the difference was only significant at the 10% level ($p = 0.086$). This may be because the breakdown message here consisted of only one word (in German) and a typo, which participants might have realized and corrected on their own. Overall, these results suggest that participants were more successful in self-repairing the breakdown following a specific (vs. generic) repair message from the chatbot.

Table 4. Results From the Analysis of Intent Recognition Scores

Breakdown type	Generic repair message		Specific repair message		p	Adjusted p	Effect size r
	M	SD	M	SD			
1 (Elaborate)	0.478	0.226	0.592	0.182	0.002	0.006	0.261
2 (Unique)	0.485	0.207	0.625	0.177	0.000	0.001	0.318
3 (Brief)	0.465	0.206	0.598	0.167	0.000	0.000	0.319
4 (Cryptic)	0.567	0.165	0.633	0.137	0.021	0.086	0.192
All	0.499	0.205	0.611	0.167			

In addition, we analyzed the impact of specific (vs. generic) repair messages on participants' evaluations of the interaction (in terms of satisfaction, frustration, and perceived chatbot effort). After confirming the reliability and validity of our measures (see Appendix B), we used *t*-tests to compare the two groups overall. We found that satisfaction was higher for participants who received specific repair messages ($M = 4.44$) than it was for those who received generic ones ($M = 4.10$; $t(577) = 3.07$, $p = 0.002$, $d = 0.255$). Their frustration levels were also lower ($M = 2.78$) than those of the participants who received generic messages ($M = 3.09$; $t(577) = -2.185$, $p = 0.029$, $d = 0.182$). Finally, participants perceived the chatbot as putting in greater effort when they received a specific message ($M = 3.69$) rather than a generic message ($M = 3.08$; $t(577) = 4.696$, $p < 0.001$, $d = 0.390$). Overall, our findings from the second design episode support the validity of our repair types and repair messages, suggesting that they would provide a solid foundation for implementing a collaborative repair strategy in InsurCo's chatbot.

Design Episode 3: Real-Time Breakdown Handling

Our third episode focused on instantiating the final repair strategy component—real-time breakdown handling—and integrating it into InsurCo's chatbot. This component comprised two key parts: (1) a breakdown type classifier responsible for automatically diagnosing breakdown types and (2) a custom web application responsible for deploying the classifier and communicating with InsurCo's chatbot backend to analyze breakdowns in real time.

Breakdown Type Classifier

To develop a breakdown type classifier that would automatically classify customer messages into one of the four breakdown types, we relied on the same content-independent message characteristics used in the cluster analysis in the first design episode: semantic weight, percentage of unknown words, word count, and words per sentence. To obtain reliable training data, we retrieved more than 1,000 randomly selected breakdown messages from customer interactions with InsurCo's chatbot that had taken place within the previous three months. We then instructed two research assistants to manually label the breakdown type of each message based on the text. They performed the initial labeling independently (intercoder agreement = 83%; Cohen's kappa = 0.80) and then resolved disagreements through discussion to arrive at a single set of labeled data for training the classifier. The labeled dataset was fairly balanced, as each category ended up with roughly the same number of messages. Cryptic-type breakdowns were the least common (150 messages), which was expected because such typically nonsensical input appeared less often in customer interactions with InsurCo's chatbot. For each message, we manually retrieved its semantic weight and percentage of unknown words, in addition to calculating the word count and average words per sentence. This dataset became the ground truth for training and evaluating the breakdown type classifier.

To identify the best-performing machine learning algorithm for our data, we compared the performance and prediction times of 10 commonly used algorithms for multiclass classification. The results, presented in Appendix C, revealed that gradient boosting performed best among the 10 algorithms. On average, it achieved an accuracy of 75.2%, a precision of 74.6%, a recall of 73.3%, and an F1 score of 73.6%. While random forest also exhibited good performance, it took more than twice as long as gradient boosting to make predictions (0.014 vs. 0.004 s). Although both numbers were small, the importance of fast prediction times led us to choose gradient boosting to train our breakdown type classifier. To identify optimal parameters for the model, we performed hyperparameter tuning using grid search with a nested 10-fold cross-validation to prevent overfitting. The best-performing model achieved an accuracy of 77.36%. We then trained the final model on all data using the identified optimal parameter values and exported it as a serialized object.

Web Application for Real-Time Breakdown Handling

To enable the chatbot to analyze and handle breakdowns in real time, we developed a web application using the Python web framework Flask. The application was designed to load our breakdown type classifier and communicate with InsurCo's chatbot backend whenever a breakdown occurred.

More specifically, it expected an HTTP request from the chatbot backend with the following request data: (1) a breakdown message, (2) the system-generated variables (semantic weight, percentage of unknown words), and (3) the types of previous breakdowns, if any. After calculating the two additional variables (word count, words per sentence), the application passed all variables to the classifier, which predicted the breakdown type and returned the corresponding value between 1 and 4. If the classifier predicted a cryptic-type breakdown, the application performed an additional check for non-German language. This allowed us to explicitly address cryptic-type breakdowns caused by foreign languages. If a language other than German was detected with confidence above 99%, the breakdown type was set to 5. Finally, the application completed the initial HTTP request by sending the prediction to the chatbot backend.

Integration Into InsurCo's Chatbot

After extensive testing of our web application, we deployed it in InsurCo's corporate cloud environment and integrated it into the running chatbot via a webhook that was triggered whenever a breakdown occurred. The webhook sent all information about the breakdown to our application and received the predicted breakdown type in return. We then configured conditional flows in the chatbot backend to enable the chatbot to respond with the corresponding repair type and repair message defined in the second design episode. This resulted in more than 20 conditional flows for all of the different combinations of breakdown types. Figure 4 provides an overview of the system architecture. Finally, we were ready to activate our collaborative repair strategy and evaluate it in the field with InsurCo's customers.

Naturalistic Summative Evaluation Through a Randomized Field Experiment

The final step of our design journey was to evaluate the instantiated collaborative repair strategy in real customer interactions with InsurCo's chatbot. Therefore, following Venable et al. (2016), we conducted a naturalistic and summative evaluation through a randomized field experiment.

Experimental Design

The field experiment employed a between-subjects design with two conditions (repair strategy: collaborative vs. non-collaborative). When breakdowns occurred in the collaborative condition, the chatbot used our newly implemented repair strategy. As detailed in the previous section, this included classifying the customer's message into one of the breakdown types and responding with a specific repair message adapted to the individual breakdown situation. In the non-collaborative condition, which served as the baseline, the chatbot used the original repair strategy that was implemented in InsurCo's chatbot before the start of our research project. This strategy put the onus on customers to resolve breakdowns by asking them to try again ("I'm sorry. I didn't understand that. Please try again."). Only after three consecutive breakdowns did the chatbot offer customers the opportunity to get in touch with a human employee (e.g., via callback). Chatbot response times were the same in both conditions so that the only difference between them was the repair strategy used.

Data and Randomization Checks

Data collection took place over an eight-week period in the fall of 2022. During this time, 4,305 customers interacted with InsurCo's chatbot and were randomized into one of the two conditions. Of these customers, 1,364 (31.68%) experienced a breakdown and thus entered the experiment. This resulted in roughly equal-sized groups of customers. We excluded 12 customers either because they closed the chat window before the chatbot could react or because there was an error in the webhook request, resulting in a final sample of 1,352 customers. For each customer-chatbot interaction, we collected conversation data (e.g., message texts), breakdown data (e.g., type, position, frequency), and customer evaluation data (satisfaction ratings, thumbs-up/down votes). In addition, we recorded information about the customer's device (mobile, tablet, desktop), web origin (advice, product detail, or order webpage), and category of request (e.g., sales, service).

To assess the efficacy of our randomization procedure, we compared the experimental groups on two basic customer characteristics (device, web origin), the likelihood of a breakdown occurring, the distribution of breakdown types, and the number of customers in each group across different time periods (day of the week, hour of the day). Chi-square test results showed that there were no systematic differences between the groups based on device ($\chi^2(2) = 1.322$, $p = 0.516$) or web origin ($\chi^2(2) = 1.989$, $p = 0.370$). Moreover, the percentage of chatbot interactions with breakdowns was similar in the two groups (32.14% vs. 31.23%; $\chi^2(1) = 0.372$, $p = 0.542$), and both groups were balanced in the distribution of breakdown types ($\chi^2(4) = 4.511$, $p = 0.341$). Finally, the results of two ANOVAs revealed that there were no significant differences in the distribution of customers to groups across weekdays ($F(1, 12) = 0.043$, $p = 0.839$) and within a day ($F(1, 45) = 0.142$, $p = 0.708$). Overall, these results indicated that the randomization was successful.

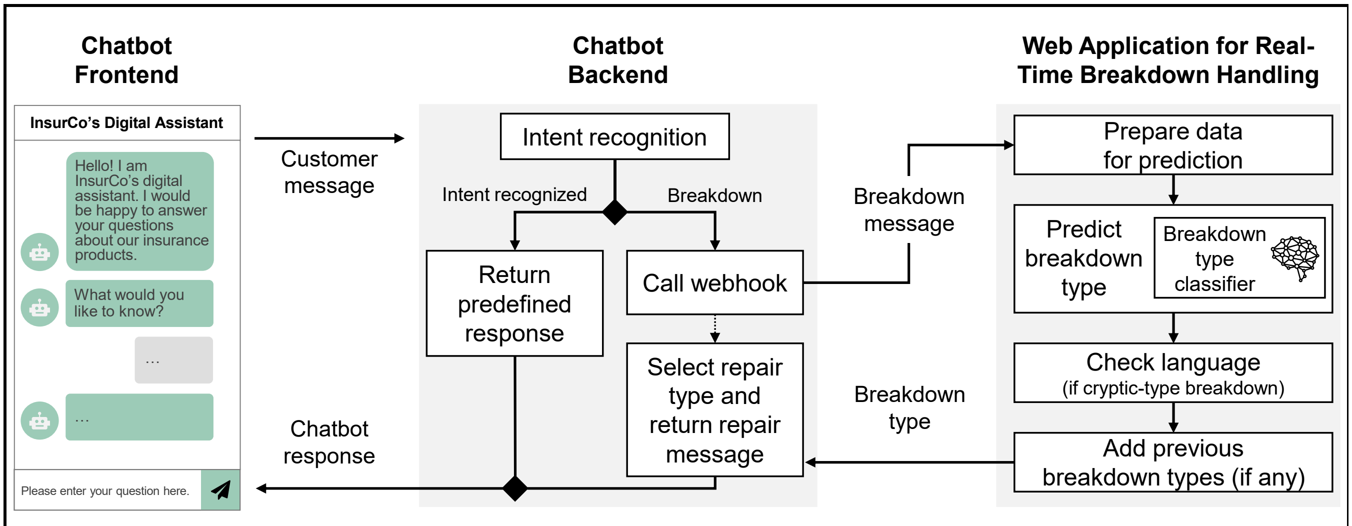

Figure 4. System Architecture

Table 5. Descriptive Statistics of Dependent Variables Across Experimental Conditions

Repair strategy	*N*	Breakdown resolution	Immediate abandonment	Customer feedback[a]	Customer satisfaction[b]
Collaborative	667	38.23%	41.68%	82.80% negative	2.07
Non-collaborative	685	32.41%	50.95%	91.39% negative	1.82

Note: [a] Based on thumbs-up/-down votes on chatbot repair messages. [b] Based on a 5-point scale (1-5 stars).

Data Analysis and Results

Guided by the two testable propositions included in our theory-driven design (see Figure 1), we conducted a series of analyses to understand the impact of a collaborative (vs. non-collaborative) repair strategy on customer interactions with InsurCo's chatbot. To test our first proposition, we used conversation data to analyze whether breakdowns were resolved (breakdown resolution) and whether customers immediately abandoned the chatbot after experiencing a breakdown (immediate abandonment). Breakdown resolution was operationalized as a binary variable indicating whether customers ultimately received a response to their request from the chatbot after the breakdown. Immediate abandonment was operationalized as a binary variable indicating whether customers abandoned the chatbot after the first breakdown or entered another message. To test our second proposition, we analyzed how customers evaluated the chatbot's repair strategy (customer feedback) and the interaction as a whole (customer satisfaction). During the experimental period, customers could provide feedback on the chatbot's first repair message using a simple thumbs-up or thumbs-down vote. Later, before closing the chat window, customers could rate their satisfaction on a 5-point scale (1-5 stars). Table 5 provides descriptive statistics for our four dependent variables.

Breakdown resolution: Whether customers were able to resolve the breakdown together with the chatbot is arguably the most important outcome for both customers and organizations, as a higher resolution rate suggests that customers were ultimately able to get the information they needed. As Table 5 shows, the breakdown resolution rate was higher in the collaborative condition (38.23%) than in the non-collaborative condition (32.41%), corresponding to a 17.96% increase in resolved breakdowns. The collaborative repair strategy was most effective in resolving elaborate- (+133.52%), brief- (+18.80%), and unique-type breakdowns (+12.41%), which together accounted for more than 90% of all the breakdowns that occurred in customer interactions with InsurCo's chatbot. In contrast, the resolution rates for cryptic-type breakdowns (Types 4 and 5) were lower in the collaborative condition (−30.05% and −38.62%, respectively), although such breakdowns were relatively uncommon (5.70% and 3.70% of the total, respectively) because they were typically caused by nonsensical input (e.g., "Iikipsg7hösöö") or languages other than German. To test the effect of a collaborative (vs. non-collaborative) repair strategy on breakdown resolution, we estimated a logistic regression model with breakdown resolution as the binary dependent variable, repair strategy as a binary independent variable, and a set of control variables (Model 1). In particular, we controlled for the customer's device, web origin, and request category, as well as the type and position of the initial breakdown in the interaction. We also included insurance product fixed effects and time fixed effects. The results in Table 6 show a significant positive effect of the collaborative repair strategy on breakdown resolution ($b = 0.243$, $p = 0.044$, odds ratio = 1.274), providing support for the proposition that a collaborative repair strategy leads to more breakdowns being resolved.

Immediate abandonment: Another critical outcome is whether customers abandon the chatbot immediately or give it a second chance, as immediate abandonment typically indicates that customers deem the chatbot incapable of ever addressing their issues. Because the majority of chatbot interactions had only one breakdown (71.60%), we analyzed the abandonment rate after the first breakdown. As Table 5 shows, the rate of immediate abandonment was lower in the collaborative condition (41.68%) than in the non-collaborative condition (50.95%), which corresponds to a 18.90% increase in customers who were willing to give the chatbot a second chance when it used a collaborative repair strategy. To test the effect of a collaborative (vs. non-collaborative) repair strategy on immediate abandonment, we estimated a logistic regression model with immediate abandonment as the dependent variable, repair strategy as the independent variable, and the same control variables as above (Model 2). Providing further support for our first proposition, the results in Table 6 show a significant negative effect of a collaborative repair strategy on immediate abandonment ($b = -0.373$, $p = 0.001$, odds ratio = 0.689), suggesting that such a strategy reduces the likelihood of customers abandoning the chatbot as soon as they experience a breakdown.

Customer feedback: To understand customers' subjective evaluations of the repair strategy, we analyzed their explicit feedback on the chatbot's repair messages (thumbs-up or thumbs-down vote). Overall, 33.51% of customers elected to offer such feedback, which was predominantly negative (87.86% thumbs down). In other words, the very occurrence of a breakdown—and the chatbot's initiation of the repair process—was poorly received by customers, regardless of the repair message. Nevertheless, as Table 5 illustrates, repair messages in the collaborative condition received fewer thumbs-down votes on average (82.80%) than those in the non-collaborative condition (91.39%), corresponding to a 9.40% decrease in customers who provided negative feedback. To test the effect of a collaborative (vs. non-collaborative) repair strategy on customer feedback, we estimated a logistic regression model with customer feedback (0 = positive, 1 = negative) as the dependent variable, repair strategy as the independent variable, and the same set of control variables as above (Model 3). The results in Table 6 show that a collaborative strategy significantly decreased the likelihood of customers providing negative feedback ($b = -1.333$, $p = 0.003$, odds ratio = 0.264). Given the potential for self-selection bias—as not all customers provided feedback—we conducted a robustness check using a two-step Heckman selection model and obtained qualitatively similar results. In line with our second proposition, these findings suggest that a collaborative repair strategy can, to some extent, reduce the negative impact of breakdowns on customers' evaluation of a chatbot.

Customer satisfaction: Finally, we analyzed customers' satisfaction ratings to understand their subjective evaluations of the entire chatbot interaction (1-5 stars). Overall, 121 customers provided a satisfaction rating (8.95%), and these ratings were rather low ($M = 1.93$, $SD = 1.40$), supporting the notion that breakdowns, in general, are a major source of dissatisfaction in customer-chatbot interaction. However, we found that average satisfaction ratings were higher in the collaborative condition ($M = 2.07$, $SD = 1.51$) than in the non-collaborative condition ($M = 1.82$, $SD = 1.30$). To test the effect of a collaborative (vs. non-collaborative) repair strategy on customer satisfaction, we estimated a linear regression model with customer satisfaction as the dependent variable and repair strategy as the independent variable (Model 4). We controlled for the customer's device, web origin, request category, and the total number of breakdowns. Again, we included insurance product fixed effects and time fixed effects. The results in Table 6 show a positive effect of a collaborative repair strategy on customer satisfaction ($b = 0.540$, $p = 0.066$, $f^2 = 0.037$), although the level of statistical significance is at the 10% level, which might be due to the relatively small sample size of customers who left a satisfaction rating. Given the potential for self-selection bias, we again conducted a robustness check using a two-step Heckman selection model and obtained qualitatively similar results. Taken together, these results suggest that a collaborative repair strategy can, to some extent, reduce the negative impact of breakdowns on customers' evaluations of the entire chatbot interaction. In summary, the findings of our naturalistic summative evaluation provide support for our two propositions by revealing that a collaborative repair strategy led to more breakdowns being resolved and mitigated the negative impacts of breakdowns on key customer outcomes.

Table 6. Effects of a Collaborative (vs. Non-Collaborative) Repair Strategy

	Breakdown resolution	***Immediate abandonment***	***Customer feedback (negative)***	***Customer satisfaction***
	Logit	**Logit**	**Logit**	**OLS**
	(1)	**(2)**	**(3)**	**(4)**
Collaborative repair strategy	0.243^{*} (0.120)	-0.373^{**} (0.115)	-1.333^{**} (0.445)	$0.540^{\dagger}$ (0.291)
Control variables	Yes	Yes	Yes	Yes
Insurance product fixed effects	Yes	Yes	Yes	Yes
Time fixed effects	Yes	Yes	Yes	Yes
Constant	−1.361 (1.282)	0.290 (1.047)	38.284^{***} (1.739)	−1.035 (1.248)
Observations	1,352	1,352	453	121

Note: The control variables are customer device, customer web origin, request category, and type and position of initial breakdown (Models 1-3 only), and total number of breakdowns (Model 4 only). Time fixed effects include a weekday-weekend dummy and time-of-day dummies. Robust standard errors are shown in parentheses. $^{***}p < 0.001$. $^{**}p < 0.01$. $^{*}p < 0.05$. $^{\dagger}p < 0.1$.

Discussion

While customer service chatbots have emerged as a promising alternative or supplement to well-established self-service applications, their imperfections make it difficult to avoid breakdowns in interactions with customers altogether. Unfortunately, the onus of resolving these breakdowns tends to fall on either the customer or the chatbot alone. Motivated by this observation, we investigated the design and impact of collaborative repair strategies that involve both the customer and the chatbot in the process of resolving a breakdown. Drawing upon TLCE-bot, we derived six meta-requirements for collaborative repair strategies and proposed a novel meta-design consisting of four components: breakdown typology, adaptive repair types, adaptive repair messages, and real-time breakdown handling. In three design episodes, we instantiated and implemented these components in InsurCo's customer service chatbot using a combination of real-world interaction data, machine learning approaches, expert feedback, and controlled online experiments. Finally, we conducted a naturalistic summative evaluation through a randomized field experiment with InsurCo's customers. Our findings suggest that a collaborative repair strategy increases customers' willingness to engage in repair, leads to more breakdowns being resolved, and mitigates the negative impacts of breakdowns on key customer outcomes. Overall, the successful instantiation of our design at InsurCo reveals that collaborative repair strategies can be implemented in a working customer service chatbot, demonstrating their real-world value and "last mile" relevance (Nunamaker et al., 2015).

Implications for Research

Our findings have several implications for research. First, they add a fresh perspective to the literature on customer-AI service interactions by shifting the narrative from confrontation to collaboration. Existing literature tends to implicitly portray customers and chatbots as adversaries. As a result, customer-chatbot interaction has generally been examined under the assumption that both sides aim to place their own interests first, minimize their own efforts, and transfer the burden of understanding to the other side. The growing body of work on chatbot breakdown and repair strategies provides a prime example of this perspective, as current repair strategies typically put the onus of resolving breakdowns on either the customer or the chatbot alone (see Appendix A). In contrast, our study introduces an alternative perspective that portrays customers and chatbots as partners, rather than adversaries, working together to resolve communication problems.

Although the idea of human-AI collaboration has gained popularity in IS research (Jain et al., 2021; Seeber et al., 2020), it has received relatively little attention in service settings, perhaps due to the inherent status differences between customers and chatbots in servant-like roles. In this regard, our perspective shift stimulates new ways of thinking, not only about means of resolving breakdowns in customer-chatbot interaction but also about the power dynamics in collaborations between humans and AI more broadly. While earlier work conceptualized AI as a tool, more recent studies often frame it as an assistant, companion, or teammate (Anthony et al., 2023; Seeber et al., 2020). Likewise, tech companies increasingly brand their AI products as "copilots" (Sacolick, 2024). As AI continues to evolve, it is essential to remain open to rethinking the power dynamics in human-AI collaboration. For example, rather than limiting our view of AI only to subordinate or equal roles (e.g., as a partner), we should broaden it to include AI taking on superordinate roles (e.g., as a leader or supervisor). This broader view could not only advance our theoretical understanding of how humans and AI work together, both today and in the future, but also inform the design of more effective human-AI collaborations.

Second, our research extends TLCE, the theoretical foundation guiding our artifact design, by integrating the perspective of customer-chatbot interaction. While prior work has argued that there is no need to restrict communication theories to interactions between humans (Guzman, 2018), our integration of TLCE with research on human-machine communication and customer service chatbots suggests that merely replacing "human" with "bot" in a theory is insufficient to account for the unique challenges in interactions with chatbots. The framework resulting from our integration, TLCE-bot, introduces three core principles that explicate the dynamics of breakdown and repair in customer-chatbot interaction and expand the scope of the original theory.

Third, our research contributes to the growing literature on breakdown and repair in customer-chatbot interaction by offering unique insights into the impact of collaborative repair strategies in real-world settings where breakdowns occur naturally. In particular, our breakdown typology improves our grasp of the nature and causes of breakdowns, while our empirical findings provide a more nuanced picture of the effectiveness of collaborative repair strategies for different breakdown types. For example, our results suggest that such strategies are effective in addressing the most common breakdown types (elaborate, unique, and brief) and thus demonstrate the overall suitability of TLCE-bot for guiding the design of repair strategies. However, collaborative repair strategies seem to struggle with the more unusual breakdowns that are caused by nonsensical input or foreign languages (cryptic), possibly because the customers are either unable to work together with the chatbot (e.g., due to language barriers) or simply do not want to (e.g., due to lack of motivation). In a broader sense, this may suggest an inherent limit to the effectiveness of (collaborative) repair strategies: establishing mutual understanding between customers and a chatbot is not always possible.

Implications for Practice

Our research offers valuable insights for practitioners involved in developing, deploying, and managing customer service chatbots, enabling them to better address breakdowns in customer-chatbot interaction and thus maximize the value of their investment in this self-service channel. With existing repair strategies, companies often face a dilemma: They want highly automated chatbots, but pushing customers into frustrating loops of trial and error can easily backfire. Our findings can help companies overcome this dilemma by providing actionable design prescriptions for collaborative repair strategies that actively involve both the customer and the chatbot in the process of resolving a breakdown. In particular, our meta-design, our artifacts, and our in-depth description of the implementation at InsurCo can serve as a blueprint for other companies to follow.

Organizations can also apply the core idea of our design to other domains where human-AI collaboration is essential, beyond customer service chatbots. Instead of leaving humans to struggle with AI-generated outputs and errors on their own, AI systems should actively involve them when failures occur or decisions need clarification. For example, if an AI-based medical diagnostic tool has low confidence in a diagnosis, it could prompt doctors for additional details about the patient's symptoms. Engaging humans in these key moments may not only enhance their understanding of how AI works but also foster more realistic expectations of its capabilities and limitations. Ultimately, this approach can enable organizations to encourage more thoughtful, reflective interactions with AI, reducing frustration and improving human-AI collaboration.

Limitations and Future Research

Our research has certain limitations that provide opportunities for future research. First, although it offers generalizable design knowledge for collaborative repair strategies, the design instantiation and evaluation focused on a specific, intent-based chatbot of an insurance company. Future studies could expand on our work by implementing our repair strategy design in large language model (LLM)-based chatbots or those with hybrid architectures. As these chatbots are prone to distraction from noisy or irrelevant context in customer requests (Shi et al., 2023), collaborative repair strategies could offer a promising approach for addressing potential misunderstandings before they lead to inaccurate responses (often called "hallucinations" or "confabulations"). Similarly, such strategies could be employed to deal with situations where policies prevent LLMs from providing a response (Wester et al., 2024). In addition, it would be valuable to explore how integrating LLMs might enhance the effectiveness of collaborative repair strategies. For example, instead of relying on manually crafted repair messages, LLMs could be used to dynamically generate tailored guidance to assist customers in resolving breakdowns even better.

Second, our typology of breakdown types originates from interactions between InsurCo's chatbot and its customers, who are mostly speakers of German. As breakdown types may manifest differently across languages, our typology should not be considered universally applicable; instead, future research should modify and expand it when investigating breakdowns in different languages. Third, future research could also draw inspiration from communication theory and cognitive psychology to address breakdowns before they occur. Unlike natural human communication, customer-chatbot interaction lacks backchanneling cues, such as gestures (e.g., nodding), vocal sounds (e.g., "mhm"), and facial expressions (e.g., a confused look). This increases communication ambiguity (Riedl, 2022) and the risk of breakdowns, as customers receive no real-time feedback on whether the chatbot understands them. Therefore, future research could explore the potential of incorporating real-time feedback capabilities inspired by natural human communication (e.g., providing linguistic feedback while customers are typing) to prevent breakdowns in the first place.

Despite its limitations, our research represents an important step toward reducing the negative impacts of breakdowns in customer-chatbot interaction. Our findings show that collaborative repair strategies can benefit both customers and organizations employing chatbots. We hope these insights will be valuable to researchers and practitioners alike and ultimately offer a new way of thinking about customer-AI service interactions.

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Appendix A

Table A1. Overview of Prior Literature on Chatbot Breakdown and Repair Strategies

Study	Research method	Theoretical foundation	Breakdown	Repair strategy	Outcomes	Main findings related to our context
(1) Breakdown exploration studies						
Dippold (2023)	Analysis of chat transcripts from 36 interactions between patients and a healthcare chatbot; interviews	Breakdown and repair in human-human communication	Breakdowns were a frequent occurrence, typically resulting from patients either (1) entering ambiguous questions and underspecified requests that the chatbot did not understand or (2) providing information requested by the chatbot (e.g., dates) in a format that it could not process. Many breakdowns occurred right at the start after patients entered verbose initial messages.	When breakdowns occurred, the chatbot responded with generic repair messages (e.g., "Sorry, I didn't understand that. Please say that again.") or offered a list of options (chatbot-driven repair). Patients tried resolving breakdowns on their own, mostly by reformulating their previous input (customer-driven repair).	Behavioral: Breakdown resolution	Breakdowns occurred frequently because questions and requests were formulated in a way that the chatbot could not understand. Patients tried resolving breakdowns on their own (e.g., by reformulating their previous input), but their efforts were often unsuccessful.
Li et al. (2020a)	Analysis of chat transcripts from 1,685 interactions between customers and a Chinese banking chatbot	N/A	Breakdowns were caused by both messages with expected content (i.e., within the scope of the chatbot's expertise) and messages with unexpected content. Most breakdowns resulted from messages with expected content that were formulated in a way that the chatbot could not understand.	When breakdowns occurred, the chatbot responded with "Sorry, I'm not sure about your meaning, please rephrase it, or leave a message on our website." After a breakdown, customers abandoned the chatbot, changed the subject, or tried resolving the breakdown by reformulating their previous input (customer-driven repair).	Behavioral: Chatbot abandon-ment	One out of four customer messages resulted in a breakdown, primarily due to ambiguous questions and out-of-scope requests. Some customers abandoned the chatbot immediately after a breakdown, while others tried resolving it on their own by reformulating their previous input.
Nguyen et al. (2021)	Analysis of chat transcripts from 451 interactions between customers and an Australasian energy provider's chatbot	Social presence theory, computers are social actors paradigm	Many interactions were affected by breakdowns, and the frequency of these breakdowns was dependent on the nature of the interaction. For example, fewer breakdowns occurred when customers inquired about general information (e.g., how to get a gas connection) than when they were seeking to resolve a specific problem (e.g., a power outage).	When breakdowns occurred, the chatbot responded with generic repair messages (e.g., "I'm sorry, I didn't understand that.") or offered a list of options ("These are some of the things I can currently help you with: …"; chatbot-driven repair). Customers tried resolving breakdowns by reformulating their input multiple times (customer-driven repair), but often did not receive the information they needed or were only referred to the company's website.	N/A	The frequency of breakdowns was influenced by the specificity of customer requests, with more specific requests being more susceptible to breakdowns. Most customers were willing to make multiple self-repair attempts.
Rapp et al. (2023)	Analysis of chat transcripts from 1,060 interactions between customers and an Italian tele-communication company's chatbot	Human-machine cooperation	Breakdowns frequently occurred because customers formulated their requests in a way that the chatbot could not understand: While some customers provided too many details about their issue or used very complex sentences, others employed a language that was too minimal for the chatbot to understand. Typographical errors were also a common cause of breakdowns.	When breakdowns occurred, the chatbot asked customers to reformulate their previous input (e.g., "Sorry, I didn't quite understand your request. Can you repeat it in a simpler way?"). Customers tried resolving breakdowns on their own using different strategies, such as adding or removing details, substituting words, or correcting typos (customer-driven repair).	N/A	Breakdowns had various underlying causes that were typically related to the way customers formulated their requests. Customers often tried to adapt their language to the chatbot's understanding capabilities, indicating their willingness to "work with" the chatbot in resolving breakdowns.
Verne et al. (2022)	Analysis of chat transcripts from 8,000 interactions between citizens and a Norwegian public agency's chatbot; interviews	Suchman's analytical framework, Taylor's theory of information needs	Breakdowns primarily occurred when requests were out of the chatbot's scope or involved long questions with multiple keywords.	When breakdowns occurred, the chatbot responded with "Please rephrase your question in simpler terms" (customer-driven repair) or offered a list of options to choose from (chatbot-driven repair). Despite providing guidance, the options often pointed to irrelevant information and complicated the path to finding a satisfactory answer.	N/A	Breakdowns were caused by out-of-scope requests and long messages that contained too many keywords for the chatbot to understand correctly. When the chatbot offered multiple potential answers, it struggled to provide the correct one within the limited set of options.

(2) Breakdown impact studies						
Brendel et al. (2023)	Controlled online experiments with university students (N_1 = 201; N_2 = 175)	Computers are social actors paradigm, social response theory, frustration-aggression theory	In the given scenario, the chatbot asked participants about their interest in joining a raffle but failed to understand them twice (regardless of their input), resulting in breakdowns.	Repair strategies were not the focus of this study, but the chatbot asked participants to repeat their request (e.g., “Sorry, I didn’t understand your response. Can you repeat it? …”; customer-driven repair).	Perceptual: Satisfaction, humanness, frustration Behavioral: Aggression	Breakdowns reduced satisfaction and perceived humanness while increasing frustration and aggression. A human-like chatbot design mitigated the intensity of aggression directed toward the chatbot.
De Sá Siqueira et al. (2024)	Controlled online experiment with participants recruited from Prolific (*N* = 258)	Technology acceptance model, social presence	In the given scenario, the chatbot asked participants about their favorite movie but failed to understand the name of the movie (regardless of their input), resulting in breakdowns.	Repair strategies were not the focus of this study, but the chatbot asked participants to reformulate their request (e.g., “Could you please try to say it differently?!? I could not understand it”; customer-driven repair).	Perceptual: Ease of use, usefulness, enjoyment, attitude, social presence	Breakdowns had negative effects on perceived ease of use, usefulness, attitude toward the chatbot, enjoyment, and social presence. A human-like chatbot communication style did not mitigate the negative effects of breakdowns.
Diederich et al. (2021)	Controlled online experiment with mostly university students (*N* = 169)	Computers are social actors paradigm, uncanny valley theory	In the given scenario, when participants requested cancelling an order, the chatbot failed to understand them twice (regardless of their input), resulting in breakdowns.	Repair strategies were not the focus of this study, but the chatbot asked participants to reformulate their request (e.g., “Unfortunately, I do not understand your request. Can you please reformulate it?”; customer-driven repair).	Perceptual: Satisfaction, humanness, uncanniness, familiarity	Breakdowns reduced perceived humanness and increased uncanniness, resulting in lower satisfaction and familiarity. A human-like chatbot design mitigated the impact of breakdowns on humanness and uncanniness.
Følstad et al. (2024)	Controlled online experiment with participants recruited from Prolific (*N* = 257)	Trust, cognitive appraisal theory, expectation confirmation theory	In the given scenario, participants used the chatbot for critical (e.g., reporting unknown transactions) and noncritical (e.g., asking about interest rates) tasks, but it failed to understand them three times (regardless of their input), resulting in breakdowns and an unsuccessful interaction.	Repair strategies were not the focus of this study, but the chatbot asked participants to reformulate their request (e.g., “I am sorry that I was not able to understand your question. Please ask again using slightly different words”; customer-driven repair).	Perceptual: Trust, emotion	Breakdowns reduced trust in the chatbot and led to negative emotional responses, regardless of the task’s criticality level. However, successful interactions after a breakdown restored trust and fostered positive emotions.
Hildebrandt et al. (2023)	Controlled online experiment with a convenience sample (*N* = 174)	Social response theory, expectation confirmation theory	In the given scenario, when participants requested changing the details of a booking, the chatbot failed to understand them twice (regardless of their input), resulting in breakdowns and an unsuccessful interaction.	Repair strategies were not the focus of this study, but the chatbot asked participants to reformulate their request (e.g., “I’m afraid I don’t know exactly what you mean. Can you say it again in a different way?”; customer-driven repair).	Perceptual: Word of mouth, anger, frustration, humanness, confirmation	Breakdowns increased frustration and anger and reduced confirmation of expectations, resulting in negative word of mouth. There was some indication that a human-like chatbot design can mitigate these effects.
Law et al. (2022)	Controlled online experiment with participants recruited from Prolific (*N* = 251)	Trust, anthropomorphism	In the given scenario, when participants asked the chatbot loan-related questions, it failed to understand them twice (regardless of their input), resulting in breakdowns.	Repair strategies were not the focus of this study, but the chatbot asked participants to reformulate their request (e.g., “I’m sorry that I was not able to understand your question. You may try to ask again using slightly different words”; customer-driven repair).	Perceptual: Trust, social presence, anthropomorphism	Breakdowns reduced trust, perceived social presence, and anthropomorphism. However, the negative effects were mitigated when the breakdowns were resolved.
Seeger & Heinzl (2021)	Controlled lab experiment with university students (*N* = 62)	Trust, anthropomorphism, attribution theory	In the given scenario, when participants asked the chatbot about their mobile phone contract, it failed to understand their question, resulting in a breakdown.	Repair strategies were not the focus of this study, and it appears that the chatbot only acknowledged the occurrence of a breakdown (“… not able to process the request …”).	Perceptual: Trust, word of mouth, anthropomorphism	Breakdowns reduced trust in the chatbot and led to negative word of mouth. A human-like chatbot design mitigated the negative impact of breakdowns on trust.

(3) Repair strategy exploration studies						
Alghamdi et al. (2024)	Literature review and analysis	N/A	Breakdowns were not the focus of this study, but misunderstandings and non-understandings are cited as the two general causes of breakdowns.	Eleven types of repair strategies were identified in the literature and grouped into two main categories. System-repair strategies were confirmation, information, solve, social, ask, and disclosure (chatbot-driven repair). User-repair strategies were clarification, information adjustment, modulation, system-centric, and user adaptation (customer-driven repair).	N/A	Existing repair strategies focus on either the chatbot or the user/ customer. Studies report mixed findings on similar strategies, and the design of repair strategies is often not grounded in theory.
Ashktorab et al. (2019)	Scenario-based online experiment with participants recruited from MTurk (*N* = 203)	Human communication theories	Breakdowns were not the focus of this study, but the challenges that chatbots face in handling the complexities of natural language are cited as the general cause of breakdowns.	Eight types of repair strategies were proposed: top, confirmation, options (chatbot-driven); ask (customer-driven); defer (involvement of human employees); keyword highlight explanation, keyword confirmation explanation, and out-of-vocabulary explanation (i.e., some collaborative elements).	Perceptual: Preferences	Although the strategies and their success rates were not examined in real chatbot interactions, there was an overall preference for repair strategy designs in which the chatbot provided a list of options, asked for confirmation, or offered explanations.
Benner et al. (2021)	Literature review and analysis	Principles of human communication: recipient design, minimization, repair	Breakdowns were not the focus of this study, but misunderstandings and non-understandings are cited as the two general causes of breakdowns.	Six types of repair strategies were identified in the literature: confirmation, solve (chatbot-driven); information, ask (customer-driven); social, and disclosure (not directly contributing to breakdown resolution). Two additions were proposed: handover (involvement of human employees) and persuasion.	N/A	The most prominent repair strategy is “ask,” where the chatbot asks users to reformulate their input. Existing strategies often include social elements (e.g., apologies) and some general information (e.g., help messages); specific guidance on how to resolve a breakdown is rare.
Feng (2023)	Literature review and taxonomy development; tests of real-world chatbots	N/A	Four issues were suggested as the underlying causes of breakdowns: language problems (e.g., foreign languages, typos), out-of-scope requests, requests that map to multiple intents, and small talk.	Repair strategies were classified in a taxonomy with five dimensions: acknowledgment, reason, explanation, repair, and handover. Most strategies acknowledge the occurrence of a breakdown but lack explanations. The most prominent strategies are “ignore the breakdown and move on” (chatbot-driven repair) and “apologize and let the user fix it” (customer-driven repair).	N/A	Repair strategies currently provide little assistance and often leave it up to the customer to resolve breakdowns. Existing strategies, particularly those implemented in real-world chatbots, offer little transparency regarding what caused a breakdown.
(4) Repair strategy impact studies						
Braggaar et al. (2023)	Controlled online experiment with a convenience sample (*N* = 150); chat transcript analysis	Human communication theories (e.g., Gricean maxims)	An analysis of 100 customer interactions with a Dutch public transportation provider’s chatbot revealed six breakdown types, three of which were selected for the experiment. Participants were shown screenshots of chatbot interactions where a breakdown had occurred due to customers (1) providing too many details, (2) providing insufficient information, or (3) asking questions beyond the scope of the chatbot.	When breakdowns occurred, the chatbot responded by either requesting a retry (customer-driven repair), offering a list of options to choose from (chatbot-driven repair), or redirecting to a human employee.	Perceptual: Satisfaction, trust, brand attitude	While no interaction effects were found between breakdown types and repair strategies, trust and brand attitude were rated higher when the chatbot escalated to a human employee or provided a list of options after a breakdown. However, asking customers to repeat their request did not result in lower satisfaction compared to the other two strategies.

Han et al. (2021)	Controlled lab experiments with university students (N_1 = 61; N_2 = 258)	Psychological reactance theory	In the given scenario, participants interacted with a chatbot to report a missing item in their delivery and request having it delivered again. Four times during the interaction, the chatbot failed to understand them (regardless of their input), resulting in breakdowns.	Study 1: When breakdowns occurred, the chatbot asked participants to try again (e.g., "I don't quite get what you're saying. Please repeat"; customer-driven repair). Study 2: The chatbot either asked participants to try again or offered a list of three predefined options to choose from (e.g., "I do not understand what you said. Can you choose one of the options below?"; chatbot-driven repair).	<u>Perceptual</u>: Service quality, satisfaction, anger, negative cognition, competence	Breakdowns increased anger and negative cognition, decreased perceived competence, and ultimately reduced service quality and satisfaction. Providing customers with a list of options mitigated the negative effects of breakdowns.
Han et al. (2022)	Controlled lab experiments with university students (N_1 = 303; N_2 = 233)	Processing fluency	In the given scenario, participants interacted with a chatbot to report a missing item in their delivery and request having it delivered again. Four times during the interaction, the chatbot failed to understand them (regardless of their input), resulting in breakdowns.	When breakdowns occurred, the chatbot asked participants to try again (e.g., "I do not understand what you said. Can you try again?"; customer-driven repair) or offered a list of three predefined options to choose from (e.g., "I do not understand what you said. Can you choose one of the options below? …"; chatbot-driven repair).	Perceptual: Service quality, satisfaction, fluency	Breakdowns decreased perceived fluency and ultimately reduced service quality and satisfaction. Providing customers with a list of options mitigated the negative effects of breakdowns.
Haupt et al. (2023)	Scenario-based online experiments with university students and participants recruited from Prolific (N_1 = 178; N_2 = 237; N_3 = 249)	Stereotype content model, attribution theory	In different scenarios, participants were asked to imagine interacting with a chatbot to seek advice about a new camera (Study 1), book a table at a restaurant (Study 2), or order pizza (Study 3). During the interactions, the chatbots failed to understand one (Studies 1 and 3) or two customer requests (Study 2), resulting in breakdowns.	When breakdowns occurred, the chatbots responded with either generic repair messages (e.g., "Sorry, I did not understand."), empathy-seeking messages (e.g., "Sorry, I did not understand. Please be patient with me as I am new to this job and … give me another chance."), or solution-oriented messages (e.g., "Sorry, I did not understand. Please try to formulate your questions or entry as precise as possible. …"; customer-driven repair).	<u>Perceptual</u>: Satisfaction, chatbot reuse intention, competence, warmth	Compared to generic repair messages, empathy-seeking (solution-oriented) messages increased perceived warmth (competence), which positively influenced satisfaction and chatbot reuse intention. The positive effect of solution-oriented messages dissolved when multiple breakdowns occurred or the chatbot (vs. customer) was blamed for the breakdown.
Huang & Dootson (2022)	Scenario-based online experiment with participants recruited from MTurk (N = 145)	Stress and coping theory	In the given scenario, participants were asked to imagine interacting with the chatbot of a fashion retailer. When questioned about the return shipping costs for an online purchase, the chatbot failed to understand the customer, resulting in a breakdown.	When the breakdown occurred, the chatbot responded with "Hmm, sorry I don't quite understand that. Please try again." (customer-driven repair). In addition, the chatbot disclosed the option of a handover at the beginning or end of the interaction: "Would you like to be connected to a live agent …?".	<u>Perceptual</u>: Aggression, problem- / emotion-focused coping	Disclosing the option to be connected to a human employee after rather than before the breakdown led customers to engage in emotion-focused coping and increased aggression.
Sheehan et al. (2020)	Scenario-based online experiments with participants recruited from MTurk (N_1 = 190; N_2 = 200)	Technology adoption, anthropo-morphism	In the scenario, participants were asked to imagine interacting with the chatbot of a hotel (Study 1) or railway company (Study 2). For specific questions about the room (Study 1) or the train ticket (Study 2), the chatbots failed to understand the customer, resulting in breakdowns.	When breakdowns occurred, the chatbots either only acknowledged them (e.g., "Sorry. I don't understand.") or asked for clarification (e.g., "Do you mean 'wi-fi'?"; chatbot-driven repair).	<u>Perceptual</u>: Adoption intention, anthropo-morphism	Breakdowns reduced perceived anthropomorphism and chatbot adoption intention. Despite the additional effort, a chatbot that asked for clarification received levels of anthropomorphism and adoption intention similar to those of a chatbot without breakdown.
Weiler et al. (2022)	Controlled online experiment with participants recruited from Prolific (N = 588)	Inoculation theory, elaboration likelihood model	In the scenario, participants interacted with a chatbot to inquire about the due date of their credit card bill. Regardless of their input, the chatbot failed to understand them, resulting in breakdowns and an unsuccessful interaction.	When breakdowns occurred, the chatbot only acknowledged them (e.g., "Sorry, I didn't get that"). However, at the start of the interaction, the chatbot used inoculation messages that contained performance information (e.g., "Our chatbot isn't perfect … it understands around 60% of user commands correctly.") and instructions ("If you encounter difficulties … rephrase your inquiry and try again"; customer-driven repair).	<u>Perceptual</u>: Discontinu-ance, trust, perceived performance	After a breakdown, higher levels of perceived performance and trust led to a decrease in chatbot discontinuance. Inoculation messages indicating a low performance level mitigated the effects of breakdowns on discontinuance.

Yeh et al. (2022)	Controlled lab experiment with a convenience sample (*N* = 126); interviews	N/A	Participants interacted with two different chatbots to perform various tasks (e.g., booking a movie ticket, searching for accommodation). On average, 25% of the messages led to a breakdown.	When breakdowns occurred, the chatbots responded with “I don’t understand your words. Could you please say it in another way?” (customer-driven repair) and provided general rules (e.g., “Avoid using abbreviations”) or examples (“If you want to find accommodation, you can say: ‘I want to find a hostel at …’”).	Perceptual: Satisfaction, preferences Behavioral: Task perfor-mance	While not all participants experienced a breakdown, those who received guidance in the form of rules rather than examples seemed to be more successful and efficient in completing tasks with the chatbot. Despite this, rule-based guidance ranked lowest among participants’ preferences.
This study	Randomized field experiment with insurance customers (*N* = 1,352); controlled online experiments with participants recruited from Clickworker.com (*N* = 579); analysis of chat transcripts from 21,736 interactions between customers and an insurance company’s chatbot	Theory of least collaborative effort, human-machine communication	Based on a cluster analysis of 5,668 breakdown-causing messages from real-world customer-chatbot interactions, four main types of breakdowns were identified: elaborate, unique, brief, and cryptic. Breakdowns primarily occurred because requests were formulated in a way that the chatbot could not understand (e.g., the request was too detailed or underspecified, too specific, or contained typos).	Both customers and the chatbot were involved in the process of resolving breakdowns: When a breakdown occurred, the chatbot diagnosed the most likely cause in real time (e.g., too many contextual details) and provided an explanation (e.g., “I struggle with long requests.”) along with specific guidance on how to resolve it (e.g., “Please shorten your request to the most important information.”). This enabled customers to self-repair the breakdown by reformulating their previous input according to the chatbot’s instructions. The chatbot’s explanations and guidance were adapted to the individual breakdown situation, taking into account the current breakdown type and any previous customer self-repair attempts rather than employing a one-size-fits-all approach.	Perceptual: Satisfaction, feedback, frustration Behavioral: Breakdown resolution, chatbot abandonment	A collaborative (vs. non-collaborative) repair strategy led to more breakdowns being resolved. It was most effective in resolving elaborate-, brief-, and unique-type breakdowns, which together accounted for more than 90% of all breakdowns that had occurred. In addition, a collaborative repair strategy reduced the rate of immediate abandonment after a breakdown and mitigated the negative impact of breakdowns on customer feedback and satisfaction.

Appendix B

Table B1. Controlled Online Experiment: Measurement Items and Reliabilities			
Construct	**Item**	**Loading**	**Source**
Satisfaction (α = 0.86, CR = 0.88, AVE = 0.71)	How satisfied are you with … … the overall chatbot interaction? … the way the chatbot treated you? … the chatbot's support?	 0.86 0.67 0.95	Verhagen et al. (2014)
Frustration (α = 0.93, CR = 0.93, AVE = 0.82)	The interaction with the chatbot … … made me feel frustrated. … made me feel annoyed. … made me feel angry.	 0.91 0.91 0.90	Gelbrich (2010)
Chatbot Effort (α = 0.92, CR = 0.92, AVE = 0.80)	The chatbot put a lot of effort into formulating its response. The chatbot worked intensively to formulate its response. The chatbot invested a lot of effort into the formulation of its response.	0.89 0.91 0.89	Tsekouras et al. (2022)

Note: All items were rated on 7-point Likert scales ranging from 1 = *strongly disagree* to 7 = *strongly agree*; α = Cronbach's alpha; CR = composite reliability; AVE = average variance extracted.

Appendix C

Table C1. Breakdown Type Classifier: Model Performance Estimation Results						
Algorithm		**Accuracy**	**Precision**	**Recall**	**F1**	**Prediction time[a]**
1	NCC	0.609 (0.005)	0.613 (0.004)	0.638 (0.005)	0.609 (0.005)	0.007 (0.0003)
2	KNN	0.735 (0.007)	0.729 (0.009)	0.711 (0.008)	0.715 (0.008)	0.009 (0.0005)
3	LR	0.742 (0.003)	0.731 (0.004)	0.715 (0.004)	0.719 (0.005)	**0.004 (0.0001)**
4	NB	0.728 (0.005)	0.718 (0.013)	0.684 (0.006)	0.685 (0.009)	0.006 (0.0003)
5	SVM	0.743 (0.006)	0.733 (0.007)	0.725 (0.007)	0.726 (0.007)	0.006 (0.0010)
6	DT	0.741 (0.008)	0.736 (0.008)	0.725 (0.010)	0.725 (0.009)	0.006 (0.0003)
7	RF	0.743 (0.008)	0.736 (0.008)	0.730 (0.008)	0.731 (0.008)	0.014 (0.0035)
8	GB	**0.752 (0.005)**	**0.746 (0.006)**	**0.733 (0.005)**	**0.736 (0.005)**	**0.004 (0.0003)**
9	AB	0.687 (0.009)	0.566 (0.034)	0.619 (0.014)	0.576 (0.020)	0.008 (0.0009)
10	NN	0.741 (0.007)	0.735 (0.009)	0.720 (0.011)	0.720 (0.012)	0.005 (0.0002)

Note: Average performance based on 30 runs as means with *SD* in parentheses. Numbers highlighted in bold represent the best performance. NCC = nearest centroid classification; KNN = *k*-nearest neighbors; LR = multinomial logistic regression; NB = naïve Bayes (multinomial); SVM = support vector machine; DT = decision tree (C4.5); RF = random forest; GB = gradient boosting; AB = adaptive boosting; NN = multilayer perceptron neural network. [a] Prediction time in seconds